

Lenovo XClarity Controller

Introduction to Lenovo XClarity Controller, the ThinkSystem server BMC

The Lenovo logo is positioned in the top right corner of the slide. It consists of the word "Lenovo" in white, sans-serif font, oriented vertically within a red rectangular background.

Lenovo

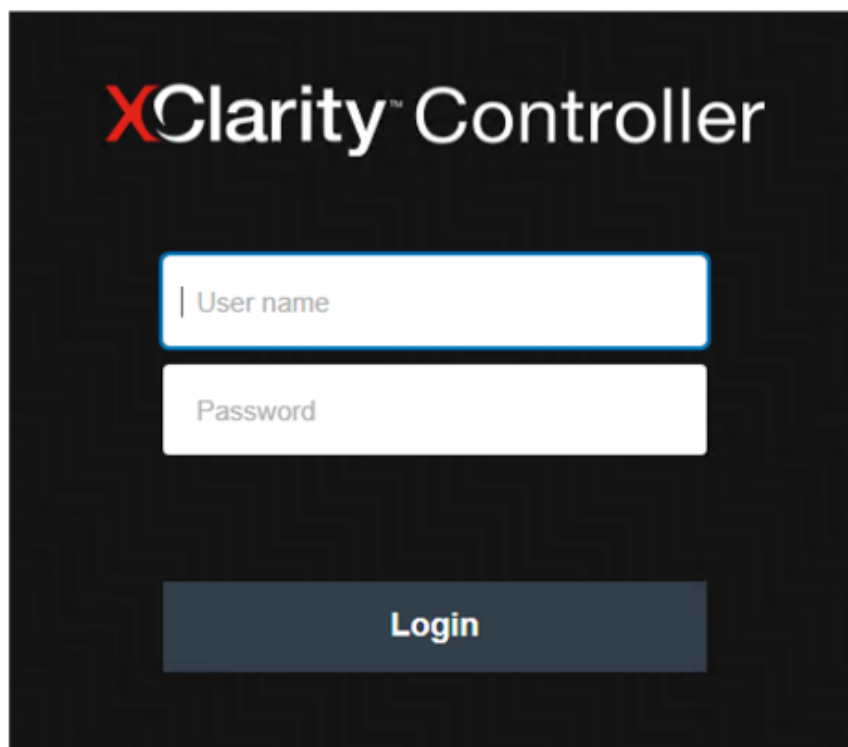
What is Lenovo XClarity Controller?

Lenovo XClarity Controller (XCC) is the baseboard management controller (BMC) for ThinkSystem servers.

XCC comes standard with all ThinkSystem servers.

XCC key features include:

- System health monitoring
- Firmware updates
- BMC setting configuration
- RAID setting configuration
- Remote control (requires an additional license key)
- Watchdog and alerting capabilities
- Event logs and service data collection
- User account management

The image shows a login interface for the XClarity Controller. It has a dark background. At the top, the text "XClarity™ Controller" is displayed in white, with the "X" in red. Below this, there are two white input fields. The first field is labeled "User name" and the second is labeled "Password". Both labels are in a small, light gray font. At the bottom of the interface, there is a dark gray button with the word "Login" in white text.

XClarity™ Controller

| User name

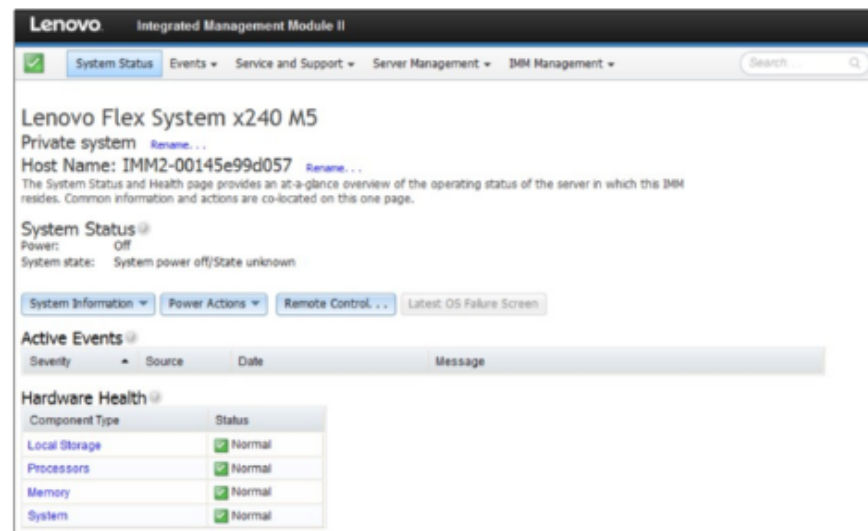
Password

Login

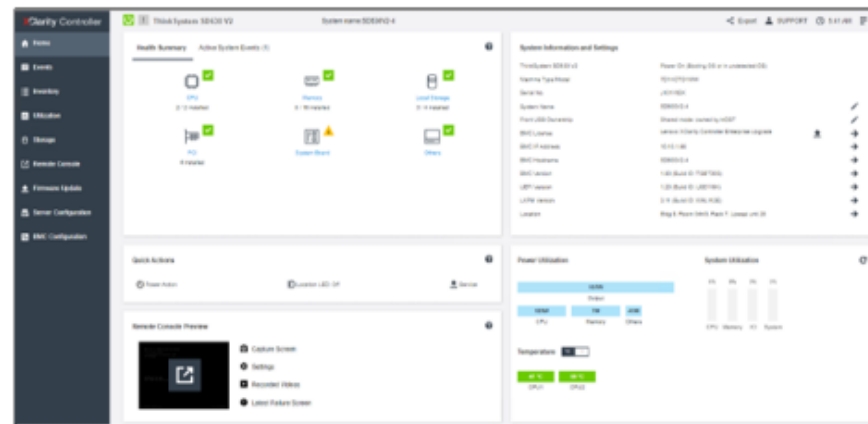
Major differences between XCC and IMM

There are a number of key differences between XCC and IMM, the BMC console for System x servers:

- The XCC GUI supports HTML5
- XCC supports National Language Support (NLS)
- Java and ActiveX clients are not supported to use the XCC remote control feature, which was replaced with HTML5 capabilities
- XCC supports international keyboard mapping
- XCC supports SNMPv3 Trap
- XCC supports Redfish
- XCC does not support Web Services-Management (WSMan)
- XCC supports the latest IPMI 2.0 specification



System x IMM homepage



ThinkSystem XCC homepage

XCC editions

There are three XCC editions: Standard, Advanced, and Enterprise.

The Standard edition is free and bundled with the system, so users do not need to manually install it.

To upgrade XCC from the Standard edition to the Advanced or Enterprise editions, users must buy the corresponding license and then use the license activation key in XCC to enable it. Click the buttons to see more information about each XCC edition.

Standard

Advanced

Enterprise

Note:

- To upgrade XCC to the Enterprise edition, users must first upgrade to the Advanced edition. If users do not follow this procedure, a warning message will be displayed on the XCC **BMC Configuration** → **License** page. Click [HERE](#) to see a screenshot.
- The ST50, ST50 V2, SR635, and SR655 do not support XCC.

XCC editions



XCC Standard edition

The Standard edition offers the following key features:

- System information and inventory gathering
- System status and health monitoring
- Alerts and notifications
- Event logging
- Network connectivity configuration
- Security configuration
- System firmware updates
- Server setting and device configuration
- Real-time power usage monitoring
- Server power remote control (Power on, Power off, Restart)
- Features on Demand (FoD) activation key management
- Ability to capture video display contents when an operating system hang condition is detected

XCC editions



XCC Advanced edition

The Advanced edition adds the following functionality to the Standard edition features:

- Ability to remotely watch video with graphics resolutions of up to 1920x1200 at 60 Hz with 16 bits per pixel
- Ability to remotely access the server using the keyboard and mouse from a remote client
- Ability to record and replay the video from a remote control session
- Ability to remotely deploy an operating system
- Component replacement logs
- Syslog alerting
- Ability to redirect the serial console via SSH
- Security Key Lifecycle Manager (SKLM)
- IP address blocking
- Graphical display of real-time and historical power usage data and temperature

XCC editions



XCC Enterprise edition

The Enterprise edition adds the following functionality to the Advanced edition features:

- Power usage capping
- Ability to map the ISO and image files located on the local client as virtual drives for use by the server
- Ability to mount the remote ISO and image files via HTTPFS, CIFS, and NFS
- Ability to collaborate across up to six users of the virtual console
- Virtual console chat
- Ability to capture and replay the server's boot-up video
- Ability to capture and replay the server's video information leading up to the point where the operating system might hang or crash
- Out-of-band (OOB) performance monitoring
- Ability to control the quality and bandwidth usage of the virtual console

How to access XCC

The default static IPv4 address assigned to XCC is 192.168.70.125.

XCC is initially configured to obtain an address from a DHCP server, and if it cannot, it uses the static IPv4 address. When booting up the server, the booting splash screen includes the BMC IP address that can be used to access the XCC Web interface. Click [HERE](#) to see a screenshot.

XCC also supports IPv6, but XCC does not have a default fixed static IPv6 IP address. For initial access to XCC in an IPv6 environment, you can either use the IPv4 IP address or the IPv6 link-local address. XCC generates a unique link-local IPv6 address using the IEEE 802 MAC address by inserting two octets with hexadecimal values of 0xFF and 0xFE in the middle of the 48-bit MAC address and flipping the seventh bit of the MAC address.

For example, if the MAC address is 08-94-ef-2f-28-af, the link-local address would be fe80:0a94:efff:fe2f:28af.

Note: The BMC IP address can be used to access XCC using telnet/SSH and to remotely access XCC with the IPMITool.

Setting up an XCC network connection using LXPM

After starting the server, LXPM can be used to configure the XCC network connection settings. The server with XCC must be connected to a DHCP server, or the server network must be configured to use the XCC static IP address. Work through the following procedure to set up the XCC network connection.

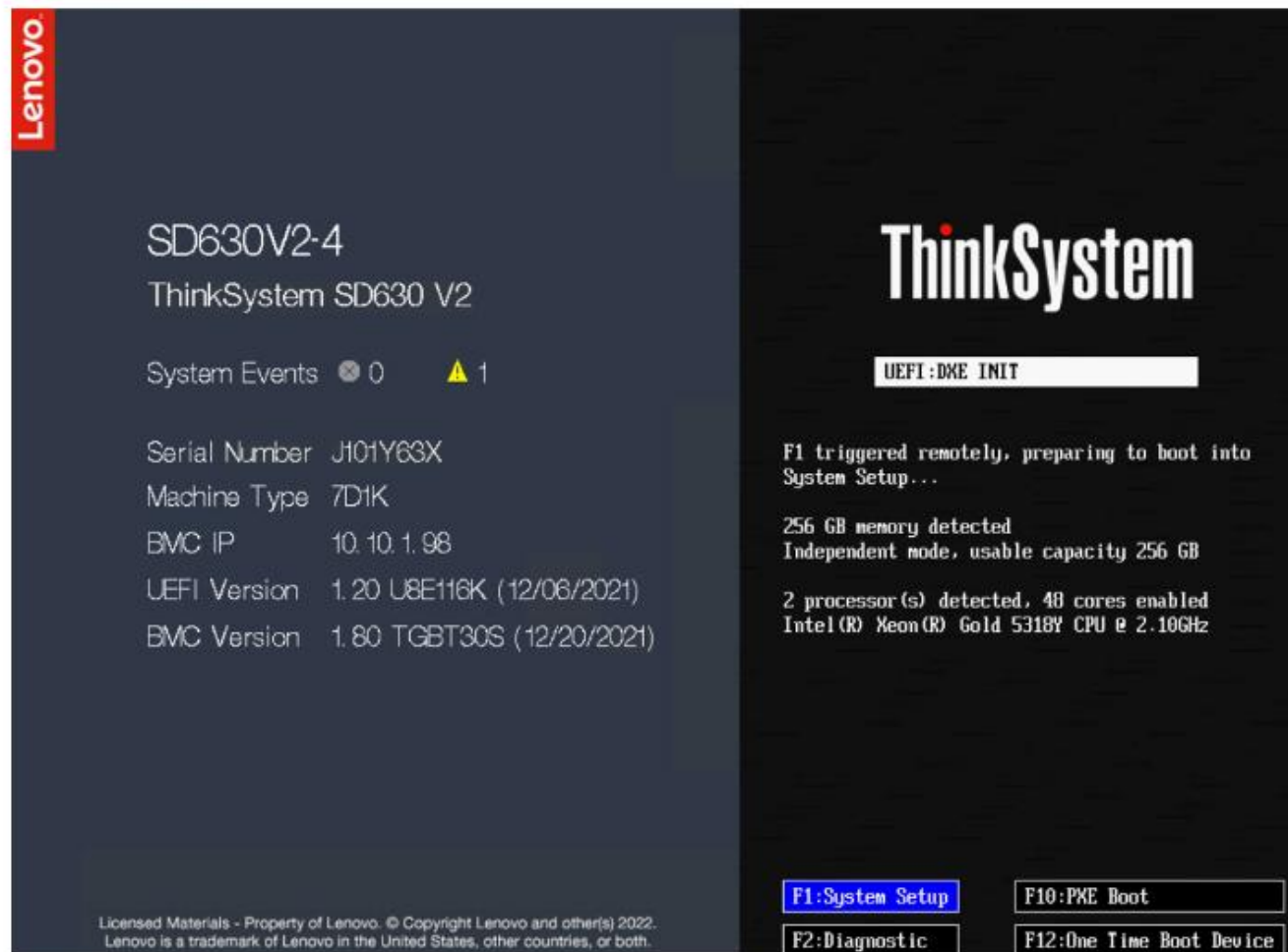
Click each number in turn to see the procedure.

Step

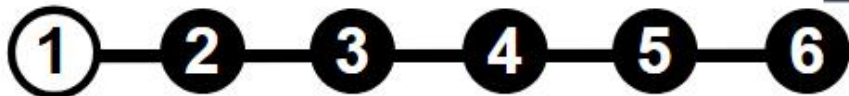


Setting up an XCC network connection using LXPM

Open LXPM by pressing F1 when you see the splash screen.



Step



Setting up an XCC network connection using LXPM

In LXPM, select **UEFI Setup** from the navigation menu.

XClarity Provisioning Manager

ThinkSystem SD630 V2

XClarity Provisioning Manager

XClarity Provisioning Manager provides an easy-to-use interface for setting up your server. After you click Apply or Skip, this page will not show again. You can access it anytime from the "?" icon at upper right corner.

Note: 1. For maximum runtime integrity, run a full memory test prior to putting a server into production. 2. Only US keyboard is applicable for correct output.

Basic System Settings

System Date: 2022 05 12 First Boot Device: VMware ESXi

System Time: 08 08 03 Boot Mode: UEFI Mode

Language: English

Management Network Basic Configuration

Network Interface Port: Dedicated Port Host Name: SD630V2-4

IP Address: 10 .10 .1 .98 Subnet Mask: 255.255.252.0

Default Gateway: 10 .10 .0 .3

BMC Credentials

Current User Name: USERID

New User Name: Current Password:

New Password: Confirm Password:

Apply Skip

Step



Setting up an XCC network connection using LXPM

Select **BMC Settings**, and then click **Network Settings** in the content area.

The image shows two screenshots of the XClarity Provisioning Manager (LXPM) interface for a ThinkSystem SD630 V2. A red arrow points from the first screenshot to the second.

First Screenshot: The left sidebar shows the navigation menu. A red box labeled '1' highlights the 'BMC Settings' option. The main content area shows 'Ethernet over USB interface' set to 'Enabled'. A red box labeled '2' highlights the 'Network Settings' link in the content area.

Second Screenshot: The 'BMC Settings' page is displayed. The left sidebar shows the navigation menu with 'BMC Settings' selected. The main content area shows various network settings: Network Interface Port (Dedicated), Fail-Over Rule (None), Burned-in MAC Address (7C-8A-E1-D6-41-69), Hostname (SD630V2-4), DHCP Control (DHCP with Fallback), IP Address (10.10.1.98), Subnet Mask (255.255.252.0), Default Gateway (10.10.0.3), IPv6 (Enabled), Local Link Address (FE80:0000:0000:0000:7EBA:E1FF:FED6:4169/64), and VLAN Support (Disabled). A red warning message at the top states: 'Attention: Must click the "Save Network Settings" at the bottom of this page to save any change on this page and its subpage.' A 'Save' button is visible on the right side of the page.

Step 1 — 2 — 3 — 4 — 5 — 6



Setting up an XCC network connection using LXPM

On the Network Settings page, select one of the **Network Interface Port** options. The BMC provides the choice of using a dedicated systems-management network connection or one that is shared with the server.

XClarity Provisioning Manager

ThinkSystem SD630 V2

Attention: Must click the "Save Network Settings" at the bottom of this page to save any change on this page and its subpage.

Network Interface Port: **Dedicated**

Fail-Over Rule

Burned-in MAC Address: 7C-8A-E1-D6-41-69

Hostname: SD630V2-4

DHCP Control: DHCP with Fallback

IP Address: 10.10.1.98

Subnet Mask: 255.255.252.0

Default Gateway: 10.10.0.3

IPv6: Enabled

Local Link Address: FE80:0000:0000:0000:7E8A:E1FF:FED6:4169/64

VLAN Support: Disabled

> Advanced Setting for BMC Ethernet

Back

Save

Discard

Default

Step



Setting up an XCC network connection using LXPM

In the **DHCP Control** field, choose between **Static IP**, **DHCP Enabled**, or **DHCP with Fallback**. This field affects what IP address is assigned to the BMC interface and whether it will be provided by DHCP or be a user-specified fixed IP address.

If **Static IP** is selected, the user must specify the IP address, subnet mask, and default gateway.

Step



Setting up an XCC network connection using LXPM

Click **Save Network Settings** to apply the changes, and then exit LXPM. It will take approximately one minute for the changes to take effect.

XClarity Provisioning Manager

ThinkSystem SD630 V2

Exit UEFI Setup

System Information

System Settings

Date and Time

Start Options

Boot Manager

BMC Settings

User Security

Network Interface Port: Dedicated

Fail-Over Rule: None

Burned-in MAC Address: 7C-8A-E1-D6-41-69

Hostname: SD630V2-4

DHCP Control: DHCP with Fallback

IP Address: 10.10.1.98

Subnet Mask: 255.255.252.0

Default Gateway: 10.10.0.3

IPv6: Enabled

Local Link Address: FE80:0000:0000:0000:7E8A:E1FF:FED6:4169/64

VLAN Support: Disabled

> Advanced Setting for BMC Ethernet

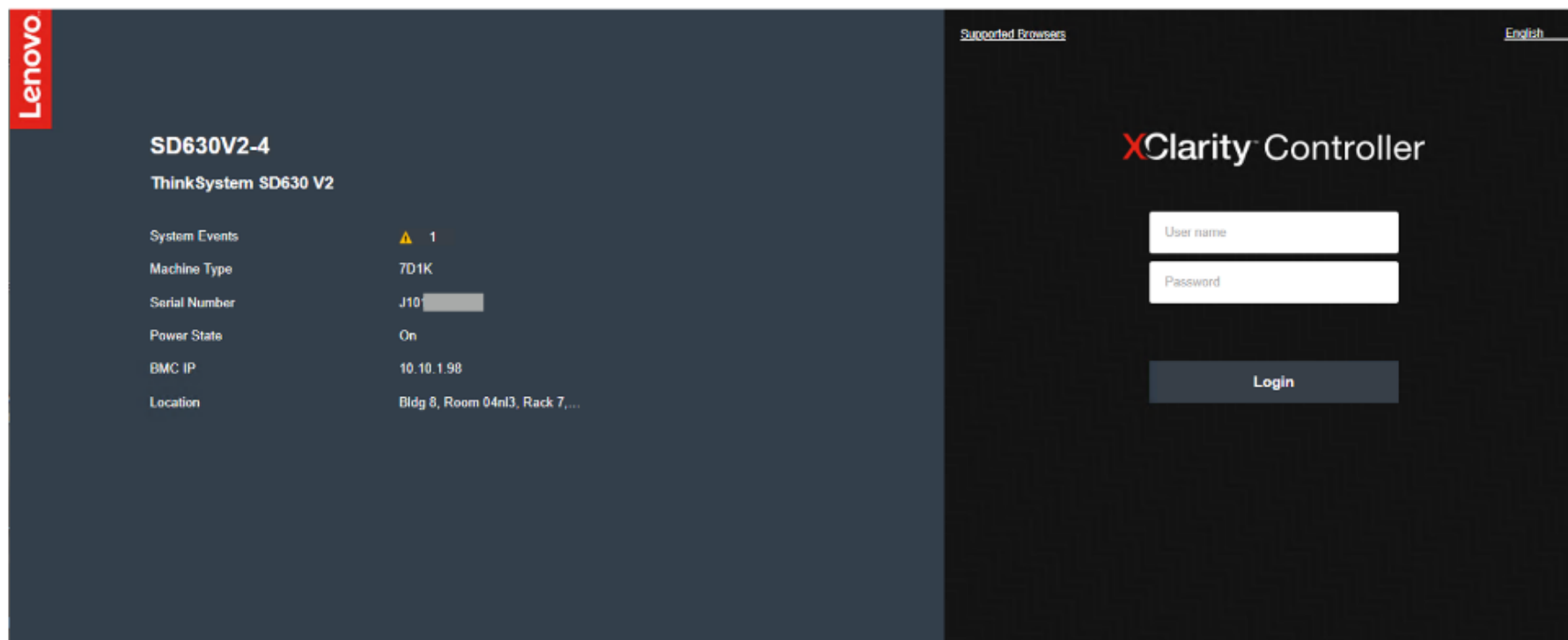
Save Network Settings

Commit the changes to BMC. Please allow a few minutes for the changes to take effect.

Step



Logging in to XCC



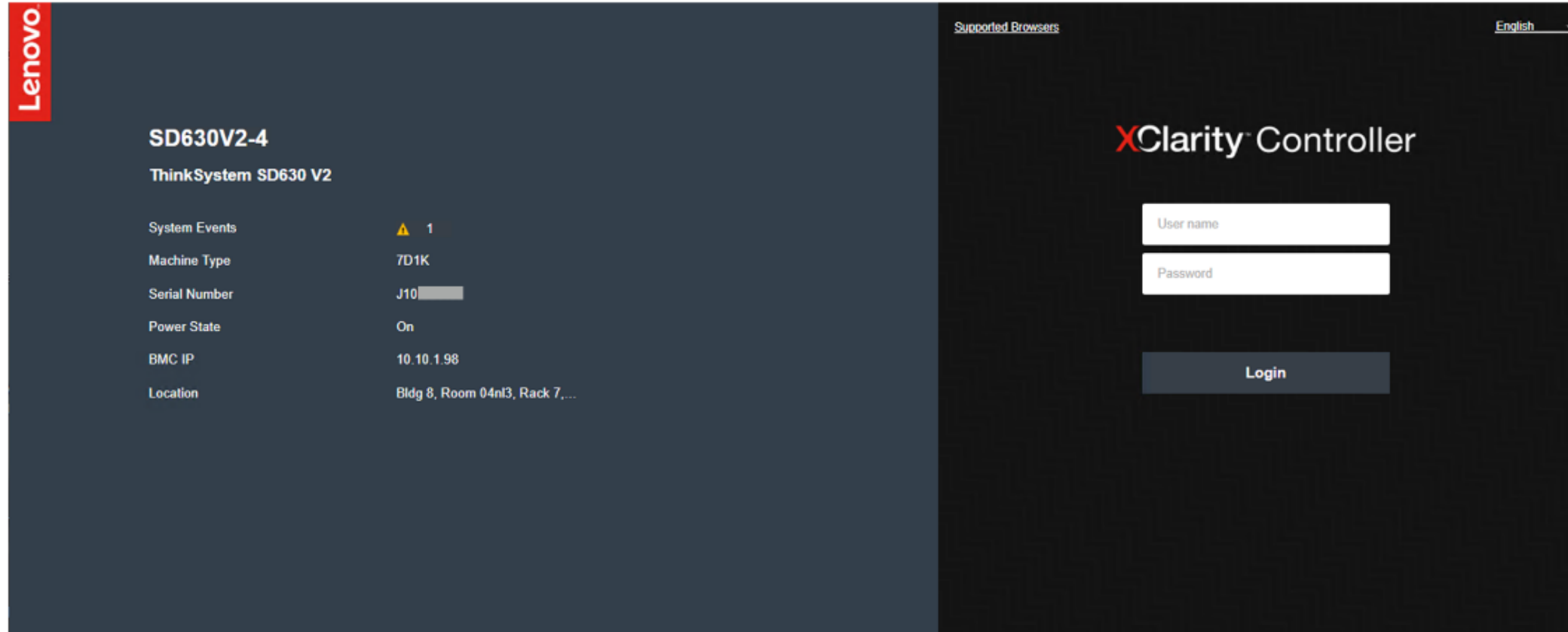
Click each number in turn to see the procedure.

Step



Logging in to XCC

Open a Web browser and type in the IP address or XCC host name.

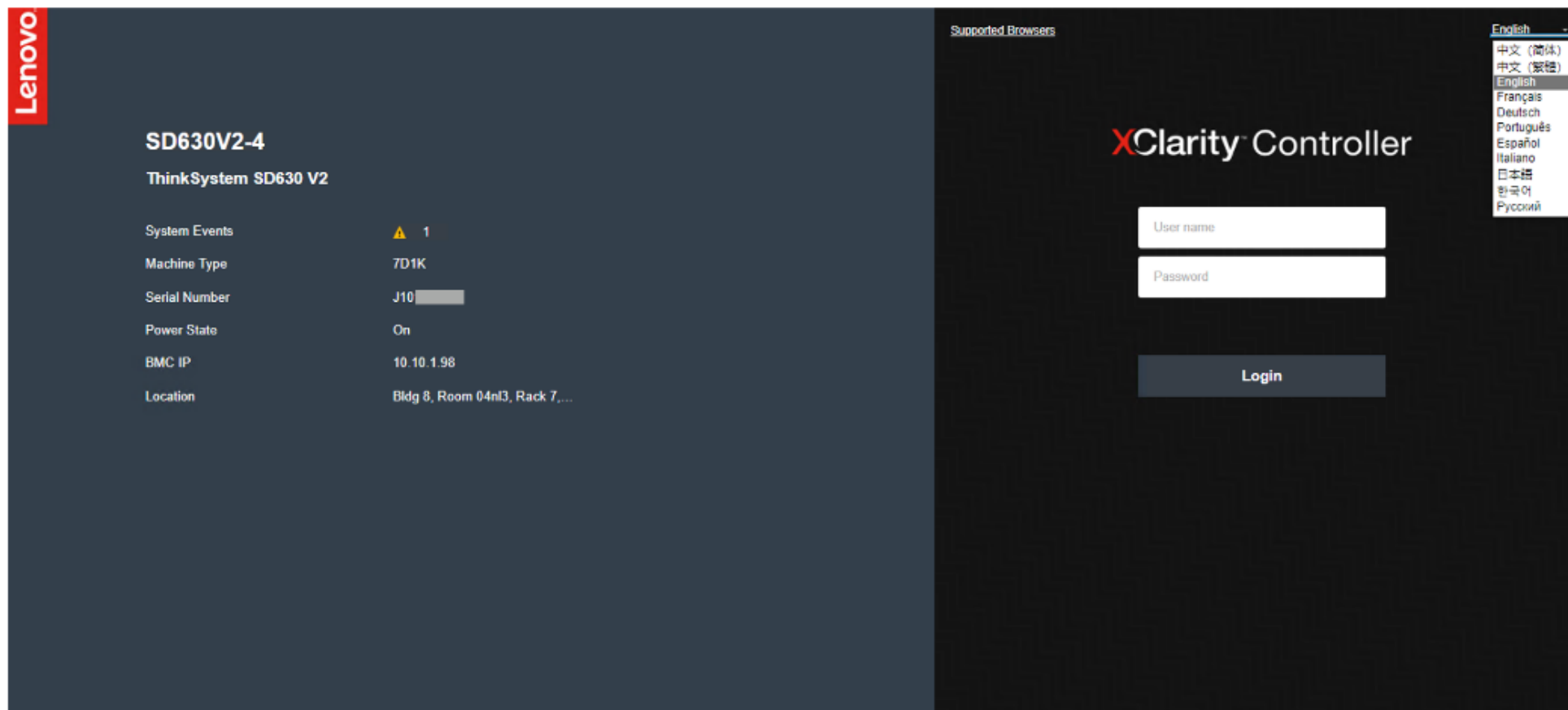


Step **1** — **2** — **3** — **4**



Logging in to XCC

Select a language from the drop-down list.



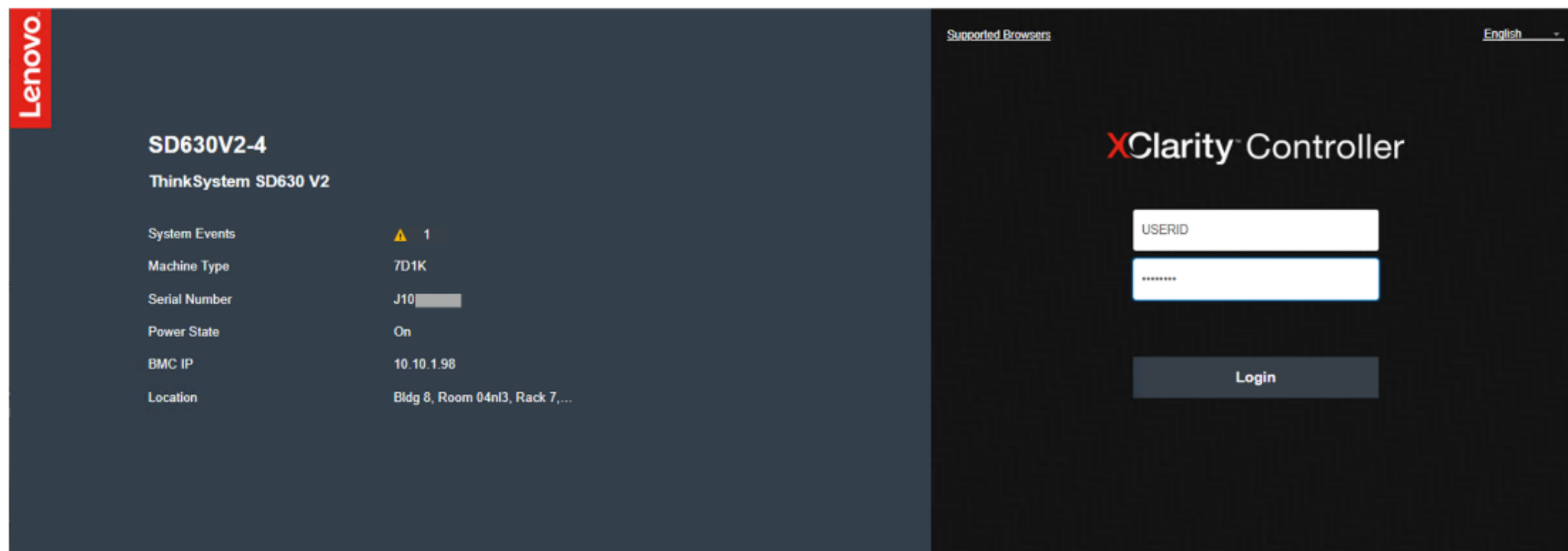
Step



Logging in to XCC

Type in the username and password. The default username is **USERID**, and the default password is **PASSWORD** (with a zero, not the letter O).

For better security, change the username and password during the initial configuration.

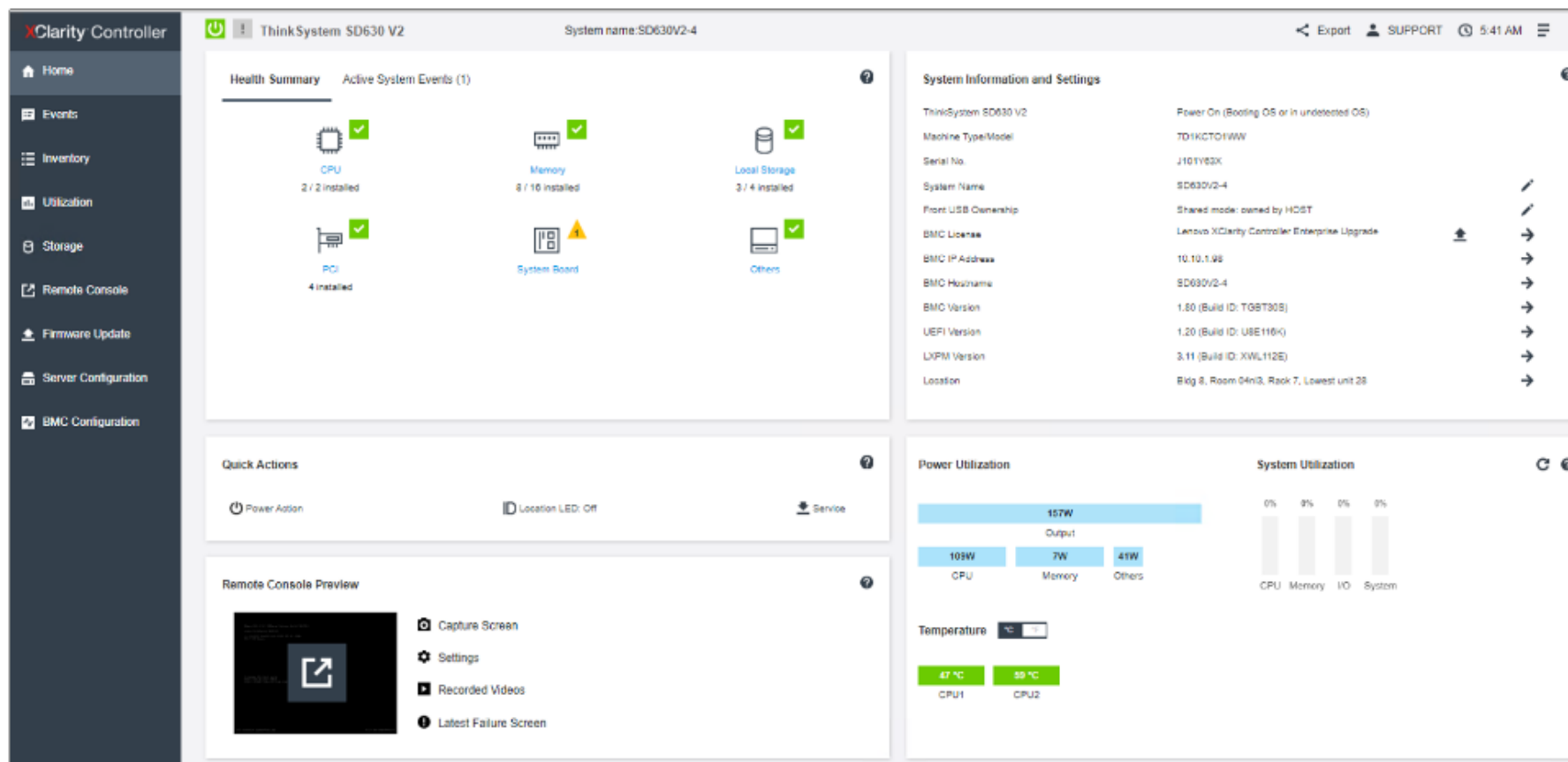


Step



Logging in to XCC

Click **Login** to start the session. The browser will open the XCC homepage.



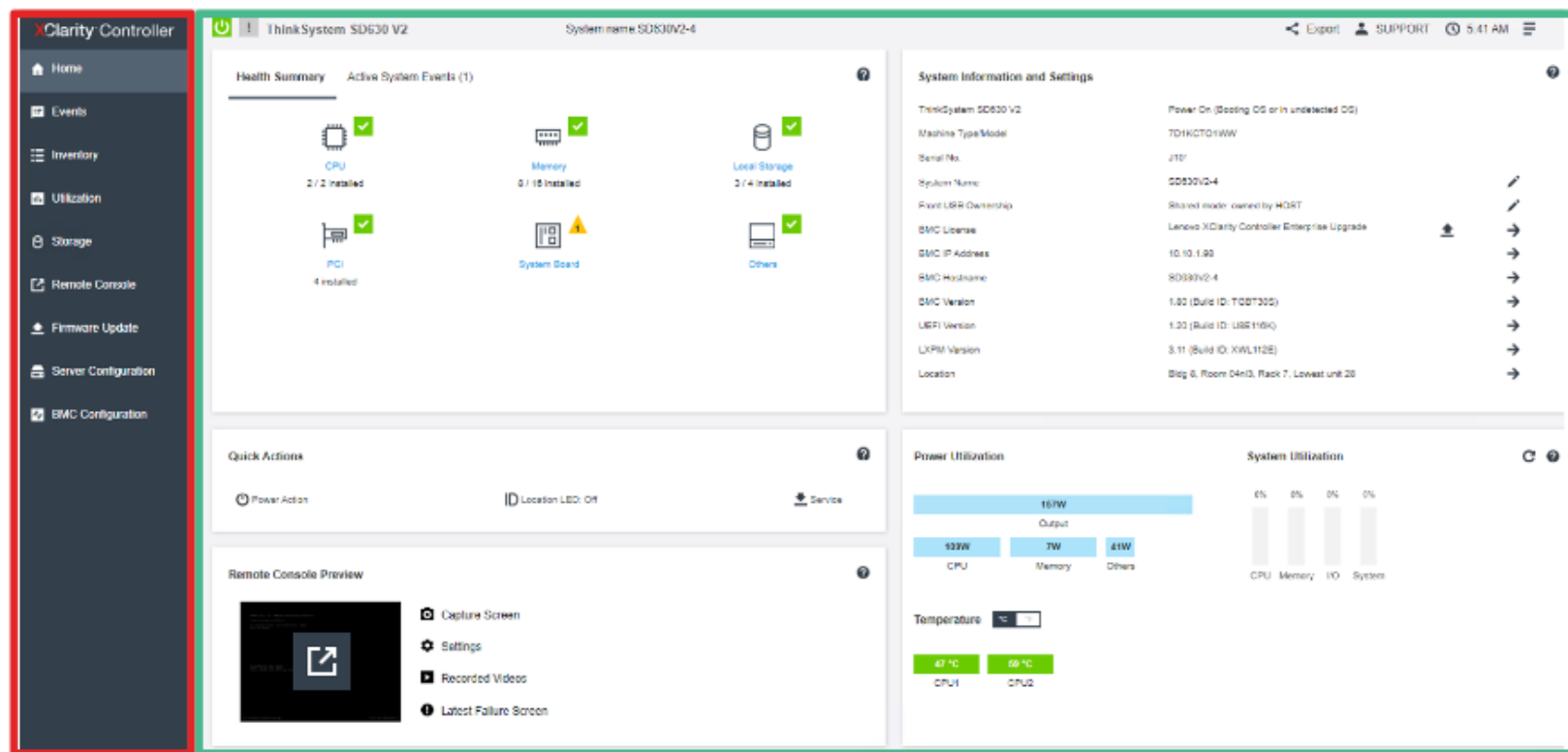
Step



XCC GUI overview

The XCC GUI is divided into two sections: the navigation panel, which lists possible actions, and the content window on the right.

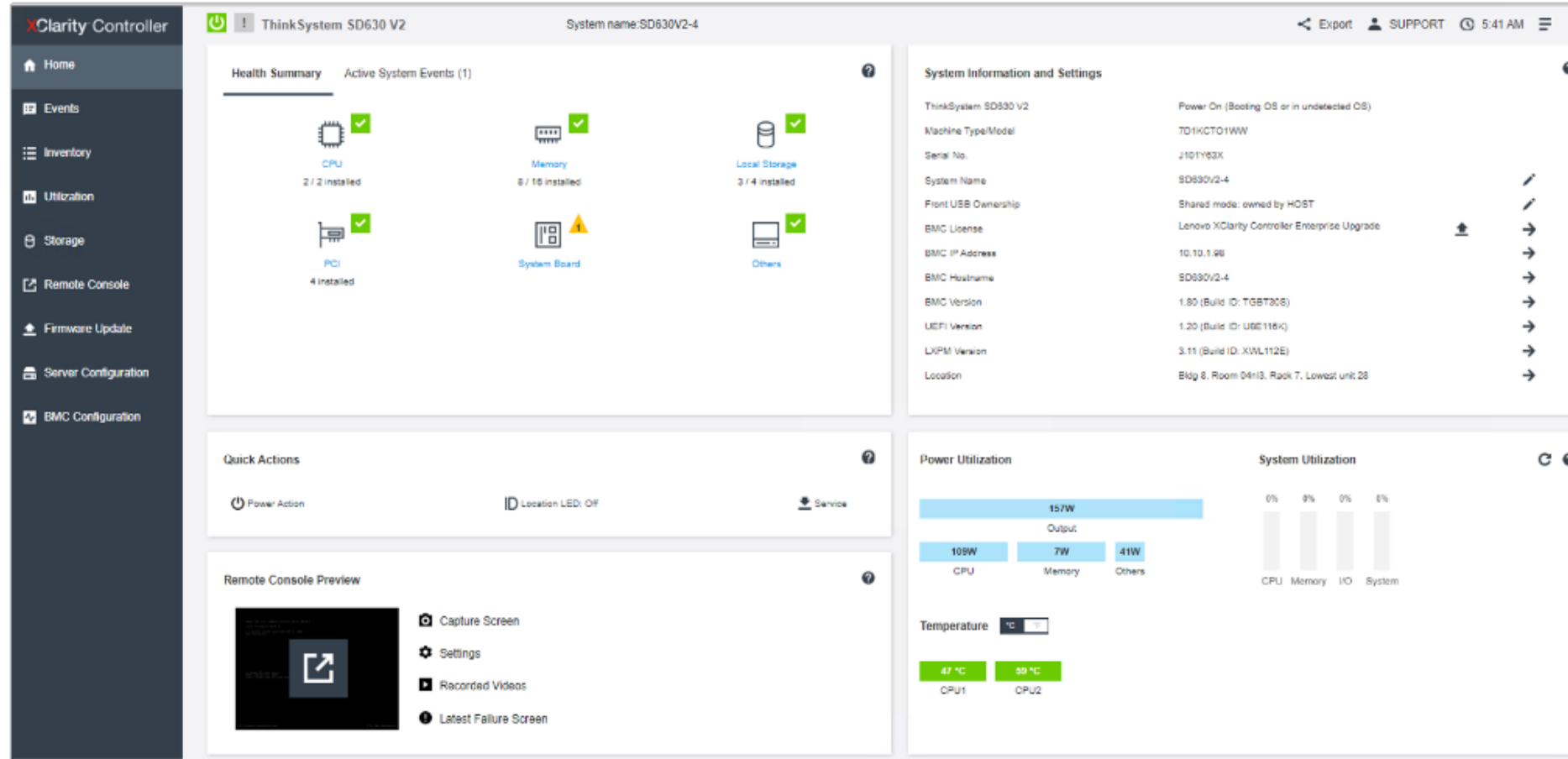
The content window uses a modular layout to give users a quick view of the server status and further actions that can be performed.



Navigation panel

XCC Homepage

The **Homepage** dashboard contains common system status information and actions.



Click here to see more information.

XCC Homepage

The **Health Summary** section contains common system status information. If any warning (yellow) or error (red) icons are displayed, users can click the icon to see more information.

The screenshot displays the XCC Health Summary interface for a ThinkSystem SD630 V2. The system name is SD630V2-4. The Health Summary section is active, showing the following components and their status:

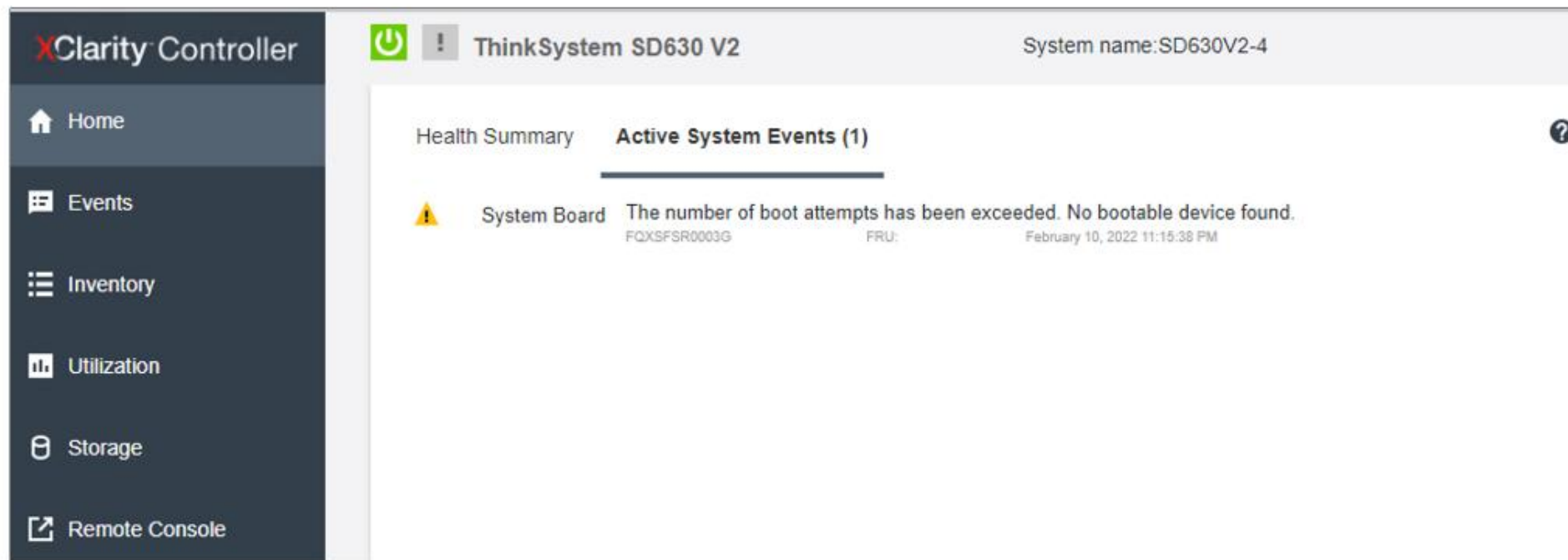
- CPU:** 2 / 2 installed (Green checkmark)
- Memory:** 8 / 16 installed (Green checkmark)
- Local Storage:** 3 / 4 installed (Green checkmark)
- PCI:** 4 installed (Green checkmark)
- System Board:** 1 warning (Yellow triangle with exclamation mark)

A red arrow points to the warning icon on the System Board. The expanded view shows the following message:

System Board The number of boot attempts has been exceeded. No bootable device found.
FQXSFSR0003G FRU: February 10, 2022 11:15:38 PM

XCC Homepage

Users can also find detailed system warning and error information in the **Active System Events** tab.



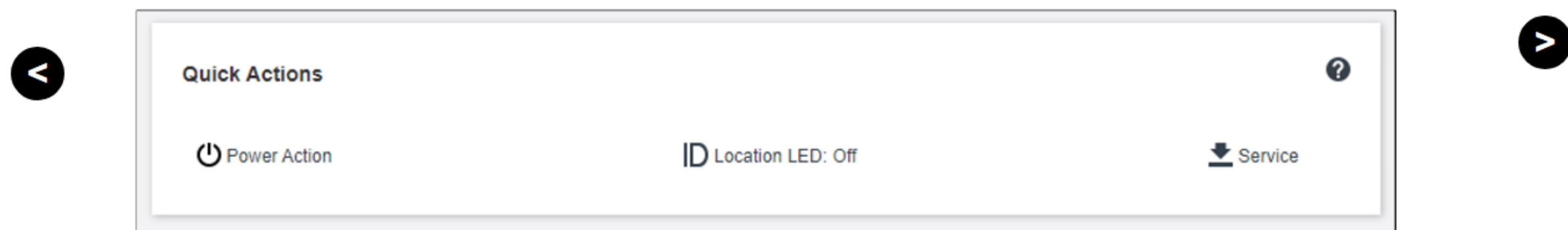
The screenshot displays the XClarity Controller interface. On the left is a dark sidebar with navigation links: Home, Events, Inventory, Utilization, Storage, and Remote Console. The main content area has a header with a power icon, a warning icon, and the text 'ThinkSystem SD630 V2'. To the right of the header, it says 'System name:SD630V2-4'. Below the header, there are two tabs: 'Health Summary' and 'Active System Events (1)'. The 'Active System Events' tab is selected and shows a warning icon, the text 'System Board', and a detailed message: 'The number of boot attempts has been exceeded. No bootable device found.' Below this message are the fields 'FRU: FQXSFSR0003G' and 'February 10, 2022 11:15:38 PM'. A help icon (?) is in the top right corner of the events section.

Component	Event Description	FRU	Timestamp
System Board	The number of boot attempts has been exceeded. No bootable device found.	FQXSFSR0003G	February 10, 2022 11:15:38 PM

XCC Homepage

The **Quick Actions** section includes the following:

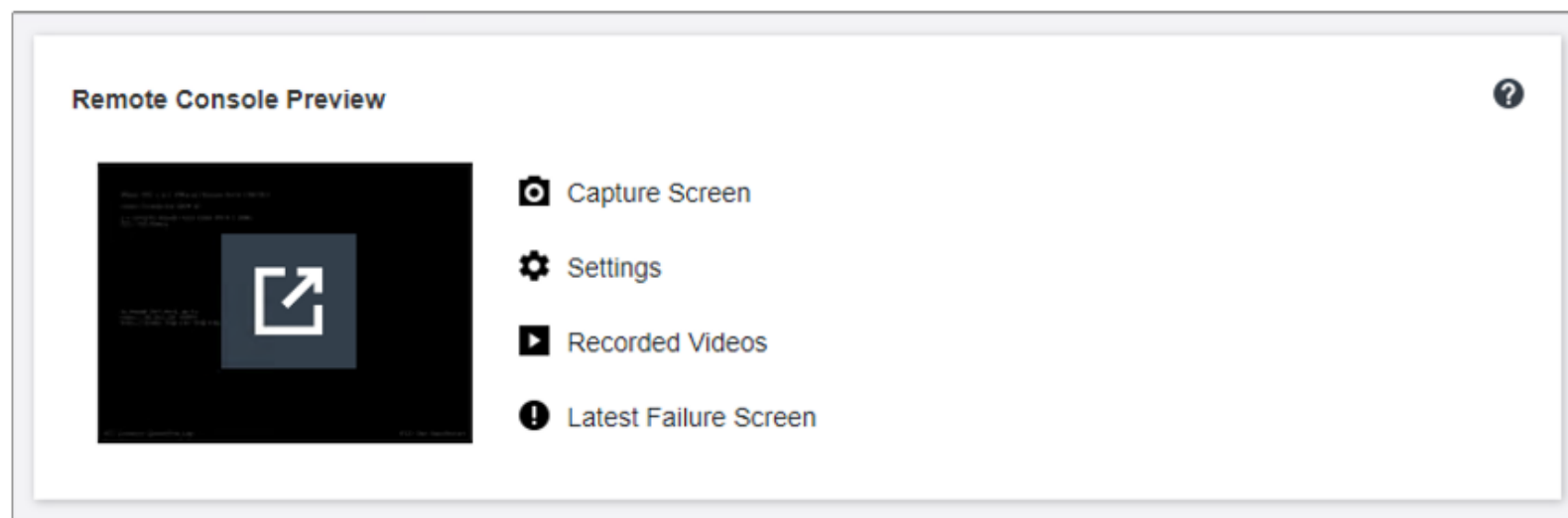
- **Power Action:** Use Power Action to control the server power or restart the server.
- **Location LED:** Use this button to control the location LED (turn on, turn off, or set to blinking) to help with locating the server.
- **Service:** Click this button to get the service data for problem determination use.



Note: The XCC service data does not include OS-level logs. To collect the complete FFDC log with OS-level logs, use OneCLI instead.

XCC Homepage

The remote console feature allows access to the server's operating system. Click the window to open a pop-up window in which users can configure virtual media and launch a remote console session on a new Web page.



Note: The remote console feature requires an XCC Advanced edition license.

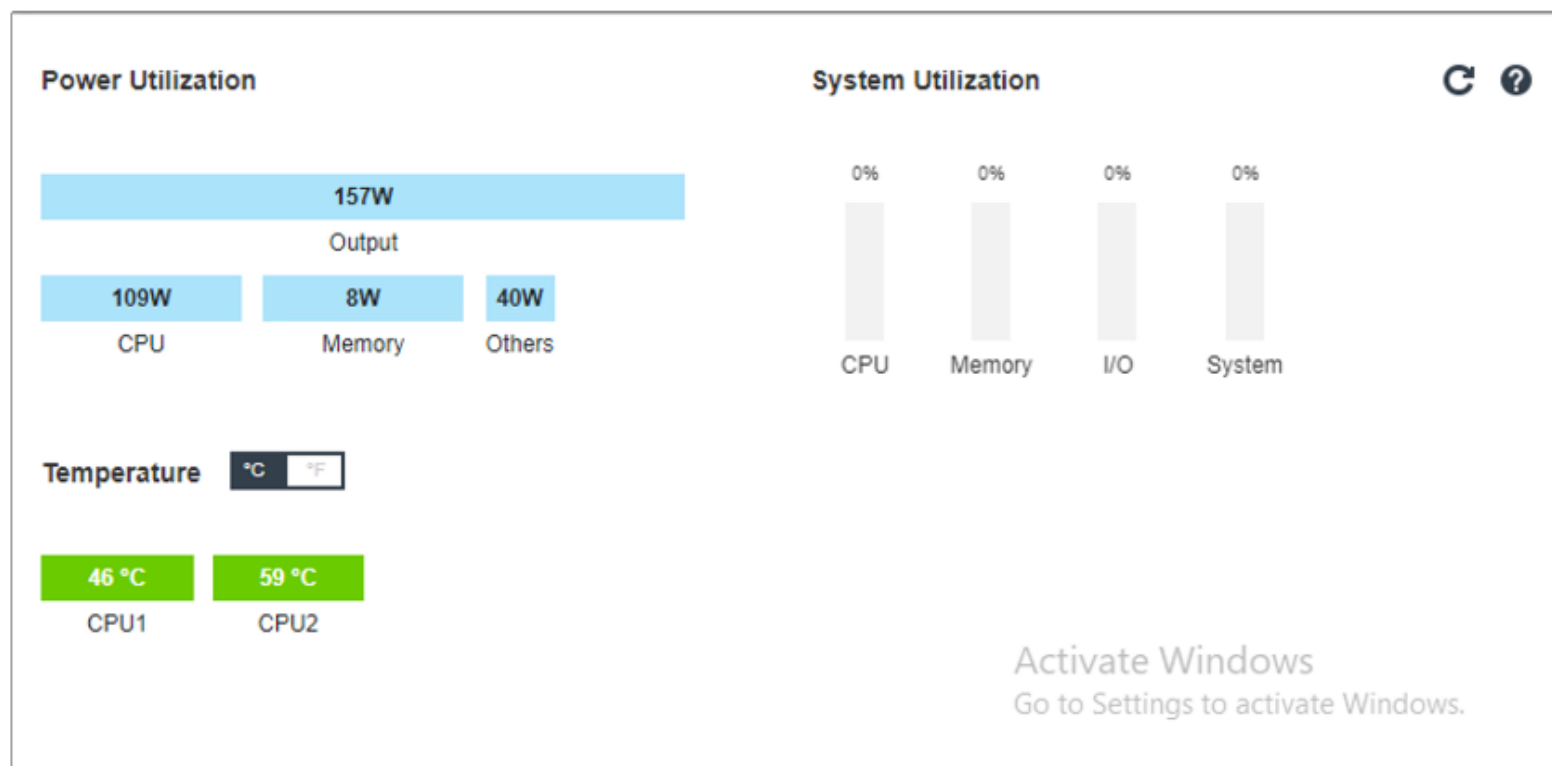
XCC Homepage

The **System Information and Settings** section contains a summary of common system information. Click the pen icon  to change the property or setting. Click the arrow icon  to see additional information. Click the upward-pointing arrow icon  to upgrade the XCC license.

System Information and Settings			?
ThinkSystem SD630 V2	Power On (Booting OS or in undetected OS)		
Machine Type/Model	7D1KCTO1WW		
Serial No.	J101Y63X		
System Name	SD630V2-4		
Front USB Ownership	Shared mode: owned by HOST		
BMC License	Lenovo XClarity Controller Enterprise Upgrade		
BMC IP Address	10.10.1.98		
BMC Hostname	SD630V2-4		
BMC Version	1.80 (Build ID: TGBT30S)		
UEFI Version	1.20 (Build ID: U8E116K)		
LXPM Version	3.11 (Build ID: XWL112E)		
Location	Bldg 8, Room 04nl3, Rack 7, Lowest unit 28		

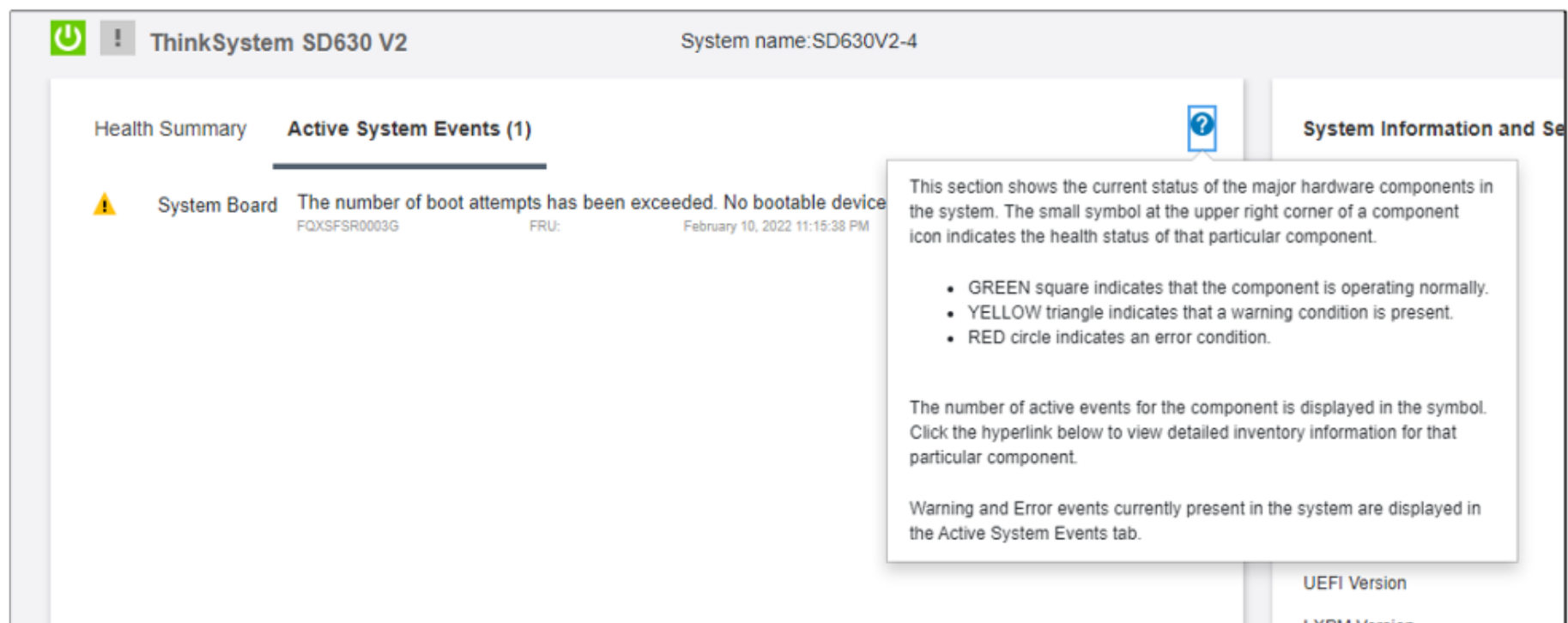
XCC Homepage

This section contains a summary of the current system power consumption, compute utilization (system, CPU, memory, and I/O subsystem), and temperature readings.



XCC Homepage

In any section, click  to see a description of the corresponding XCC function.



ThinkSystem SD630 V2 System name:SD630V2-4

Health Summary **Active System Events (1)** 

 System Board The number of boot attempts has been exceeded. No bootable device
FQXSF0003G FRU: February 10, 2022 11:15:38 PM

This section shows the current status of the major hardware components in the system. The small symbol at the upper right corner of a component icon indicates the health status of that particular component.

- GREEN square indicates that the component is operating normally.
- YELLOW triangle indicates that a warning condition is present.
- RED circle indicates an error condition.

The number of active events for the component is displayed in the symbol. Click the hyperlink below to view detailed inventory information for that particular component.

Warning and Error events currently present in the system are displayed in the Active System Events tab.

UEFI Version
LXDM Version



Events page

The **Events** page → **Event Log** tab contains a historical list of all hardware and management events.

XClarity Controller

System name: SD630V2-4

ThinkSystem SD6...

Event Log Audit Log Maintenance History Alert Recipients

Customize Table Clear Logs

Refresh

Type: [Error] [Warning] [Information] All Event Sources All Dates

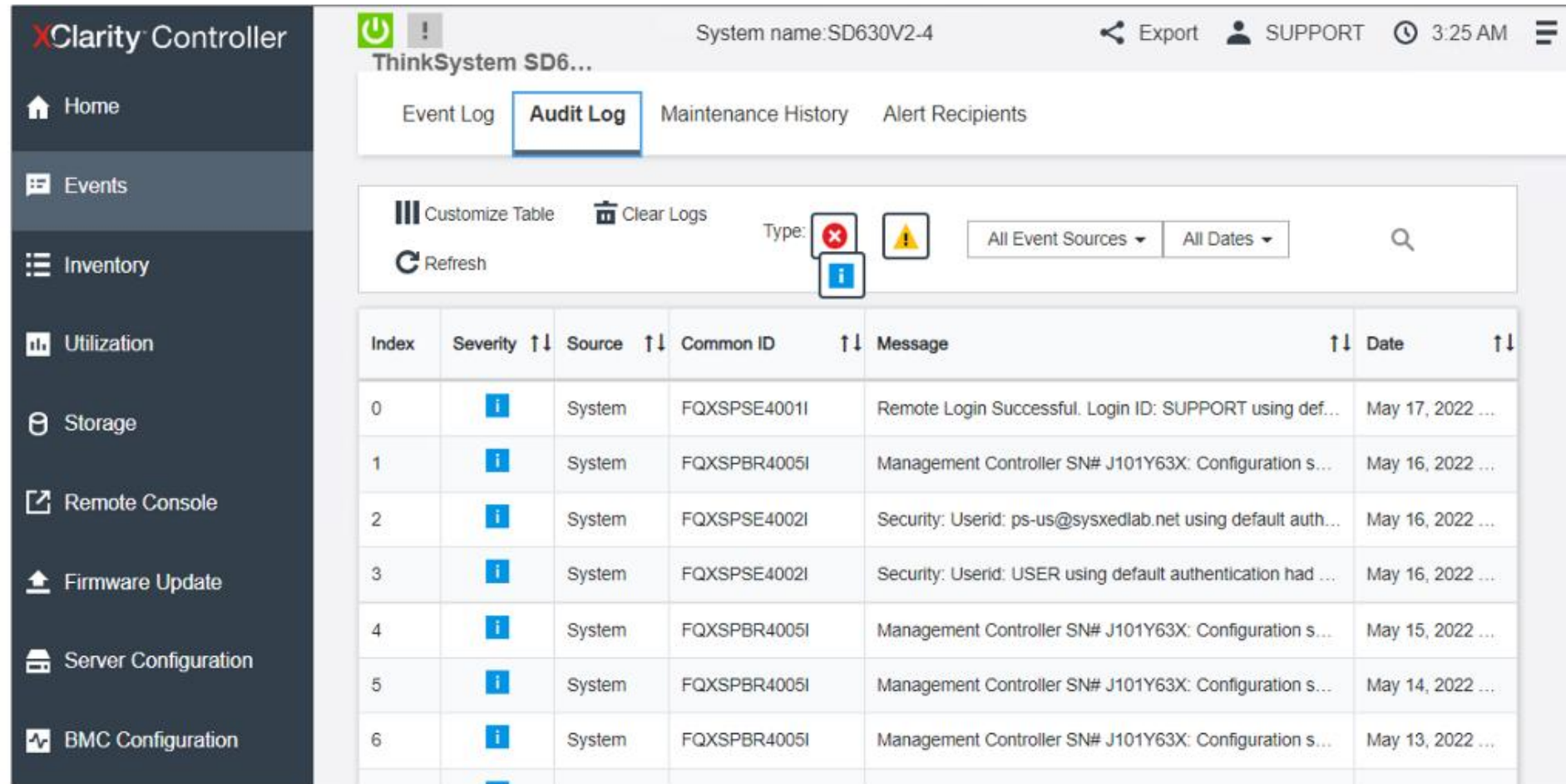
Index	Severity	Source	Common ID	Message	Date
0	[i]	System	FQXSPNM4011I	ENET[CIM:ep1] DHCP-HSTN=SD630V2-4, DN=sysxedlab.net, IP@=...	May 1
1	[i]	System	FQXSPPW0009I	Host Power has been Power Cycled.	May 1
2	[i]	System	FQXSPNM4011I	ENET[CIM:ep1] DHCP-HSTN=SD630V2-4, DN=sysxedlab.net, IP@=...	April 2
3	[i]	Power	FQXSPPW2008I	Host Power has been turned on.	March
4	[i]	Power	FQXSPPW0008I	Host Power has been turned off.	Febru
5	[i]	Power	FQXSPPW2008I	Host Power has been turned on.	Febru
6	[i]	Power	FQXSPPW0008I	Host Power has been turned off.	Febru



Click here to see more information.

Events page

The **Audit Log** tab contains a historical record of user actions, such as logging in to XCC, creating a new user, and changing a user password.



XClarity Controller

System name: SD630V2-4

Export SUPPORT 3:25 AM

ThinkSystem SD6...

Event Log **Audit Log** Maintenance History Alert Recipients

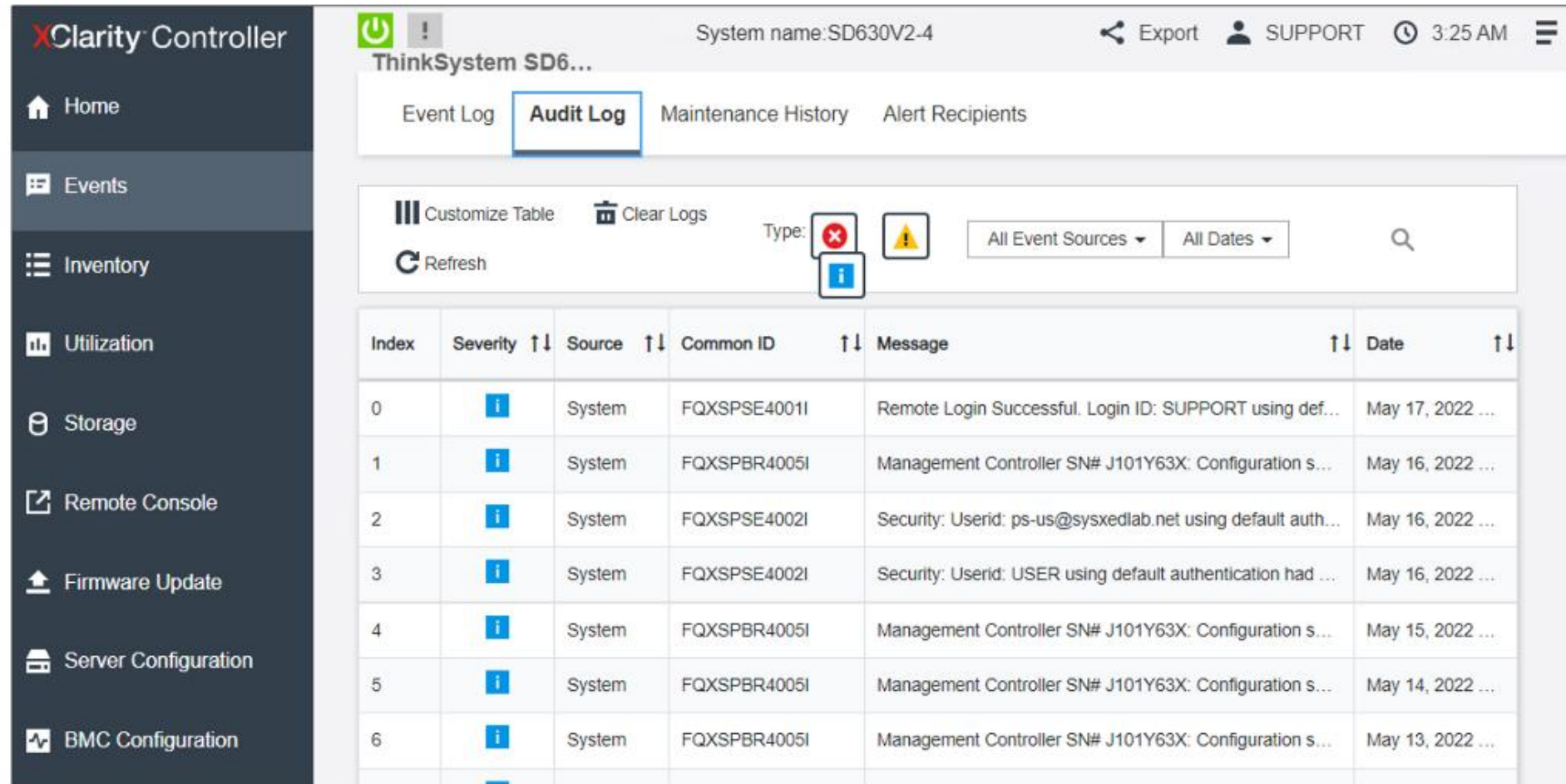
Customize Table Clear Logs Type: [Error] [Warning] [Info] All Event Sources All Dates

Refresh

Index	Severity	Source	Common ID	Message	Date
0	[Info]	System	FQXSPSE4001I	Remote Login Successful. Login ID: SUPPORT using def...	May 17, 2022 ...
1	[Info]	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 16, 2022 ...
2	[Info]	System	FQXSPSE4002I	Security: Userid: ps-us@sysxedlab.net using default auth...	May 16, 2022 ...
3	[Info]	System	FQXSPSE4002I	Security: Userid: USER using default authentication had ...	May 16, 2022 ...
4	[Info]	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 15, 2022 ...
5	[Info]	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 14, 2022 ...
6	[Info]	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 13, 2022 ...

Events page

The **Audit Log** tab contains a historical record of user actions, such as logging in to XCC, creating a new user, and changing a user password.



The screenshot displays the Lenovo XClarity Controller interface. On the left is a dark sidebar with navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration, and BMC Configuration. The main content area is titled 'ThinkSystem SD6...' and shows the 'Audit Log' tab selected. Above the table are controls for 'Customize Table', 'Clear Logs', 'Refresh', and filters for 'Type' (with error and info icons), 'All Event Sources', and 'All Dates'. The table lists audit events with the following data:

Index	Severity	Source	Common ID	Message	Date
0	Info	System	FQXSPSE4001I	Remote Login Successful. Login ID: SUPPORT using def...	May 17, 2022 ...
1	Info	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 16, 2022 ...
2	Info	System	FQXSPSE4002I	Security: Userid: ps-us@sysxedlab.net using default auth...	May 16, 2022 ...
3	Info	System	FQXSPSE4002I	Security: Userid: USER using default authentication had ...	May 16, 2022 ...
4	Info	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 15, 2022 ...
5	Info	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 14, 2022 ...
6	Info	System	FQXSPBR4005I	Management Controller SN# J101Y63X: Configuration s...	May 13, 2022 ...

Events page

Use the **Alert Recipients** tab to add and modify email and syslog notifications or SNMP trap recipients.

XClarity Controller

System name: SD630V2-4 | Export | SUPPORT | 3:32 AM

ThinkSystem SD6...

Event Log | Audit Log | Maintenance History | **Alert Recipients**

Email/Syslog Recipients (0) | + Create | SMTP Server | Retry and Delay | ⚙️

Name	Method	Address	Events to Rec	Action
Create Email Recipient				
Create Syslog Recipient				

SNMPv3 User (0) | + Create | SNMP

User	Access	Address for traps	Authentication protocol	Privacy protocol	Action
------	--------	-------------------	-------------------------	------------------	--------

Inventory page

The **Inventory** page contains components information for the system.

The screenshot shows the XClarity Controller web interface. The left sidebar contains navigation links: Home, Events, Inventory (selected), Utilization, Storage, Remote Console, Firmware Update, Server Configuration, and BMC Configuration. The main content area displays system information for 'ThinkSystem SD630 V2' (System name: SD630V2-4). It includes an 'Export' button, a 'SUPPORT' link, and a clock showing 7:18 AM. The 'Inventory' section is divided into two main components: CPU and DIMM. The CPU section shows 'CPU: 2/2 Installed' and a table with details for two CPUs. The DIMM section shows 'DIMM: 8/16 Installed' and a table with details for one DIMM. A 'Quick Link' sidebar on the right provides shortcuts to various system components.

XClarity Controller ThinkSystem SD630 V2 System name:SD630V2-4 Export SUPPORT 7:18 AM

CPU: 2/2 Installed

Socket	Model	Max Cores	Part ID
CPU 1	Intel(R) Xeon(R) Gold 5318Y CPU @ 2.10GHz	24	3330 3736 3536 3430 (30765640)
CPU 2	Intel(R) Xeon(R) Gold 5318Y CPU @ 2.10GHz	24	3330 3736 3536 3430 (30765640)

FRU Name CPU 1 **L1 Data Cache Size** 1152 KB
Manufacturer Intel(R) Corporation **L1 Instruction Cache Size** 768 KB
Max Speed 3400 MHz **L2 Cache Size** 30720 KB
Maximum Data Width Capable 64-bit NOT capable **L3 Cache Size** 36864 KB
Family Intel Xeon **Voltage** 2 V
Max Threads 48 **External Clock** 100 MHz

DIMM: 8/16 Installed

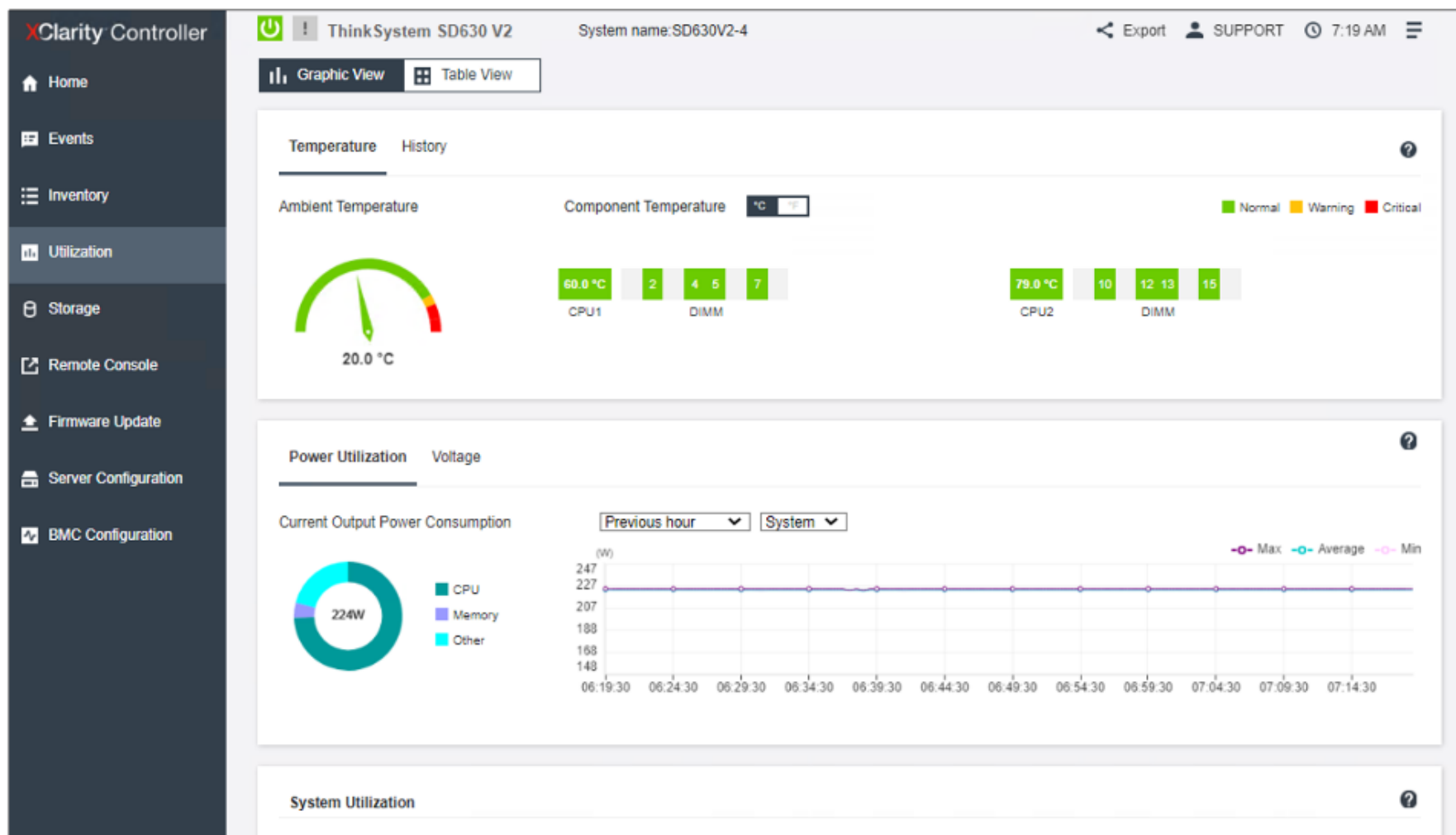
Slot	Type	Capacity	Part Number
DIMM 2	DDR4	32 GB	M393A4K40EB3-CWE

Quick Link

- CPU
- DIMM
- DISK
- PSU
- Fan
- PCI
- SYS Board
- Others
- SYS FW

Utilization page

The **Utilization** page contains a summary of common server utilization information.



Click the image to switch to a table view.

Storage page

The **Storage** → **Detail** page contains detailed drive information.

XClarity Controller

Home

Events

Inventory

Utilization

Storage

Detail

RAID Setup

Remote Console

Firmware Update

⏻

!

ThinkSystem SR860 V2

System name:

Export

USERID

6:07 AM

⋮

Storage Detail

Controller 1: (PCI Slot 15)

Location	Manufacturer	Product	Status	Capacity	Interface	Media	Form Factor	Miscellaneous
Bay 1	Intel	SSDSC2KB24	✓ Normal	240 GB	SATA	SSD	2.5"	100% Remaining Life >
Bay 3	Intel	SSDSC2KB24	✓ Normal	240 GB	SATA	SSD	2.5"	100% Remaining Life >

Backplane

Name	Drive Bays	Manufacturer	Serial Number	Part Number	FRU Number
Backplane 1	Bay 0 - Bay 7	LNVO	XXXXX9CD0019	SC57A26300	



Click here to see more information.

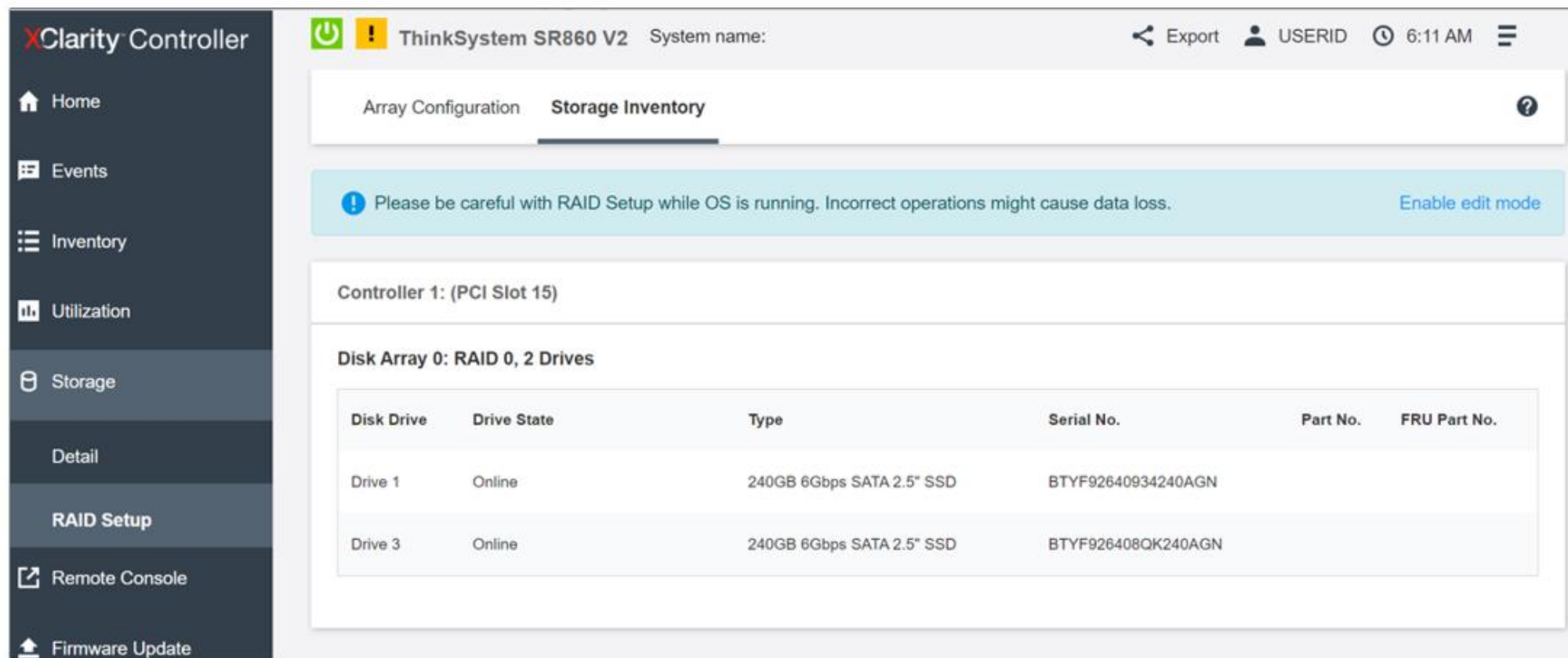
Storage page

The **Storage** → **RAID Setup** page can be used to configure RAID settings. At present, the RAID Setup page cannot be used to set up the RAID configuration for NVMe drives.

The screenshot displays the XClarity Controller interface for a ThinkSystem SR860 V2 system. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage (selected), Detail, RAID Setup, Remote Console, and Firmware Update. The main content area is titled 'Array Configuration' and 'Storage Inventory'. A warning message states: 'Please be careful with RAID Setup while OS is running. Incorrect operations might cause data loss.' Below this, 'Controller 1: (1 virtual disk created)' is shown. A 'RAID 0' configuration is displayed for 'Virtual Disk 1', indicating an 'Optimal' status and a capacity of '445.172GB'. The configuration is associated with 'Disk Array 0, RAID 0'. A 'Controller Actions' dropdown menu is visible on the right side of the controller section.

Storage page

The **Storage** → **RAID Setup** → **Storage Inventory** tab contains disk array and drive information for the system.



XClarity Controller

ThinkSystem SR860 V2 System name: Export USERID 6:11 AM

Array Configuration **Storage Inventory**

Please be careful with RAID Setup while OS is running. Incorrect operations might cause data loss. [Enable edit mode](#)

Controller 1: (PCI Slot 15)

Disk Array 0: RAID 0, 2 Drives

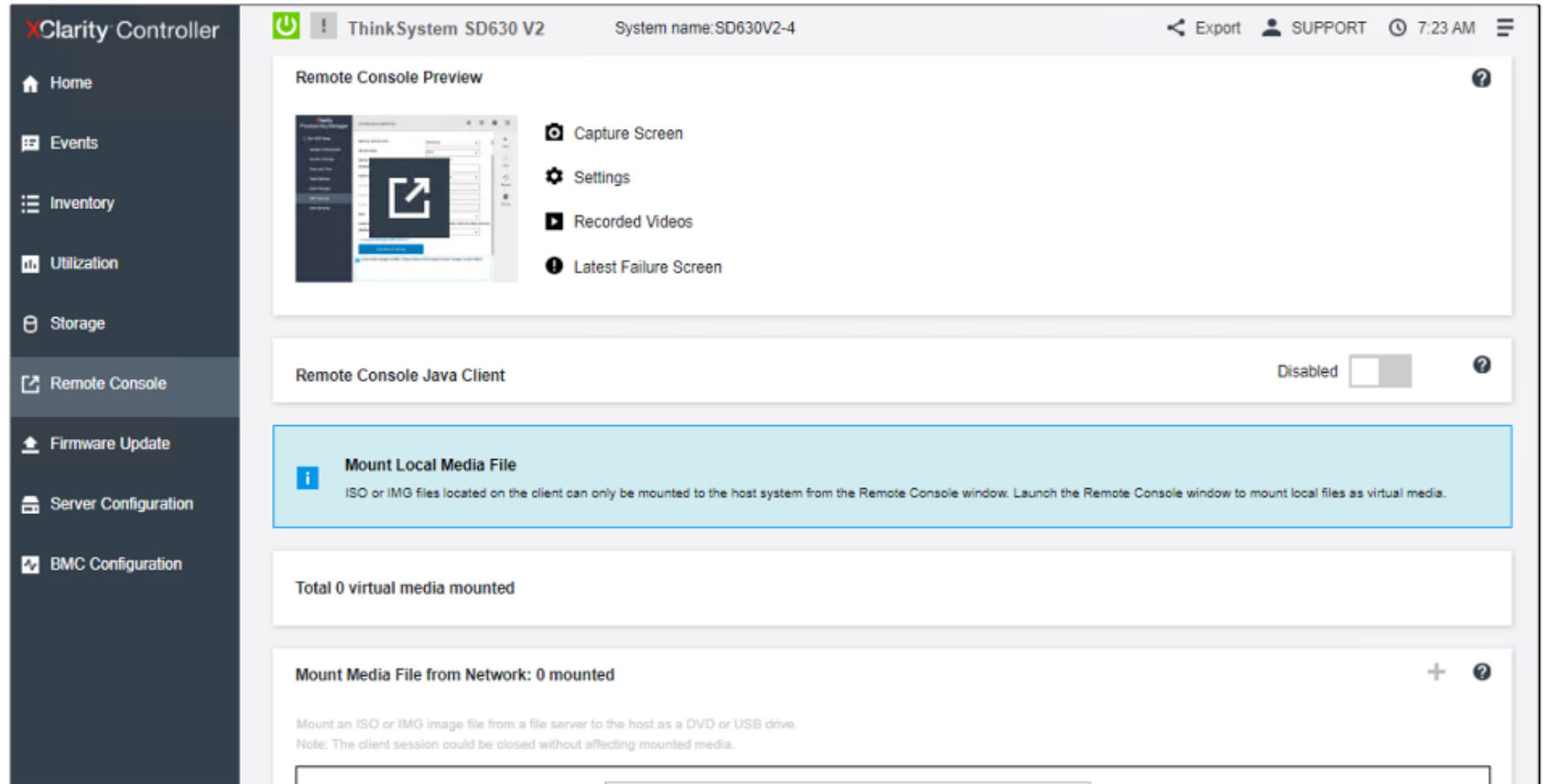
Disk Drive	Drive State	Type	Serial No.	Part No.	FRU Part No.
Drive 1	Online	240GB 6Gbps SATA 2.5" SSD	BTYF92640934240AGN		
Drive 3	Online	240GB 6Gbps SATA 2.5" SSD	BTYF926408QK240AGN		

Remote Console page

Use the **Remote Console** page to remotely control the system or mount local media files to the remote system.

The **Remote Console Preview** feature requires an XCC Advanced edition license.

The **Mount Local Media File** feature requires an XCC Enterprise edition license.



Firmware Update page

Use the **Firmware Update** page to update system firmware. Some component firmware updates might require XClarity Essentials tools (Update Xpress or OneCLI).

The screenshot shows the 'Firmware Update' page in the XClarity Controller interface. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, **Firmware Update** (selected), Server Configuration, and BMC Configuration. The main content area is titled 'ThinkSystem SD630 V2' with 'System name:SD630V2-4'. It features three sections: 'System Firmware', 'Adapter Firmware', and 'Update from Repository'. The 'System Firmware' section contains a table with columns: Type, Status, Version, Build, and Release Date. The 'Adapter Firmware' section includes a note about updating via BMC and a table with columns: Slot No., Device Name, Status, Version, Manufacturer, and Release Date. The 'Update from Repository' section has a dropdown menu set to 'CIFS' and an 'Input URL' field.

System Firmware

Type	Status	Version	Build	Release Date
BMC (Primary)	Active	1.80	TGBT30S	2021-12-20
BMC (Backup)	Inactive	1.41	TGBT20G	2021-07-26
UEFI	Active	1.20	USE116K	2021-12-08
LXPM	Active	3.11	XWL112E	2021-11-09
LXPM Windows Drivers	Active	3.11	XWL312E	2021-12-20
LXPM Linux Drivers	Active	3.11	XWL212E	2021-12-20

Adapter Firmware

This section lists the adapter firmware that can be updated via the BMC. For adapters not listed here, Lenovo XClarity Essentials can be used to update firmware.
Note: please wait until the system boots into F1 Setup or OS.

Slot No.	Device Name	Status	Version	Manufacturer	Release Date
OnBoard	Mellanox ConnectX-4 Lx 10/25GbE LOM	Active	14.31.1014	Mellanox Technologies	2021/09/29

Update from Repository

CIFS Input URL:

Firmware Update: Update from Repository

The **Firmware Update** page has an **Update from Repository** section. Users can create a repository, upload firmware files to the repository, and then use the XCC Firmware Update page to update the system firmware from the repository.

The repository can be a common Internet file system (CIFS – previously known as SMB) mount or a network file system (NFS) mount.

The screenshot shows the 'Update from Repository' section of the XClarity Controller interface. On the left is a dark sidebar with navigation links: 'Storage', 'Remote Console', 'Firmware Update' (highlighted), 'Server Configuration', and 'BMC Configuration'. The main content area has a title 'Update from Repository' with a help icon. Below the title is a form with a dropdown menu currently showing 'CIFS' (with 'CIFS' also highlighted in blue). To the right of the dropdown are four input fields: 'Input URL:', 'User Name:', 'Password:', and 'Domain:'. Below these fields is a 'Mount Options:' label followed by an input field. At the bottom of the form are three buttons: 'Connect', 'Update', and 'Reset'.

Field	Value
File System	CIFS
Input URL:	
User Name:	
Password:	
Domain:	
Mount Options:	

Buttons: Connect, Update, Reset

Server Configuration page

The **Server Configuration** → **Adapters** page contains information about the adapters installed in the server.

The screenshot shows the XClarity Controller interface for a ThinkSystem SD630 V2 server. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration (highlighted), Adapters (highlighted), Boot Options, Server Properties, and BMC Configuration. The main content area displays the adapter status for Mellanox ConnectX-4 Lx 10/25GbE LOM (OnBoard). A blue information box at the top states: "This page displays the status information for adapters that support status monitoring. To check out the inventory information of all installed adapters, visit the [Inventory](#) page." Below this, the adapter details are shown: Product Name: Mellanox ConnectX-4 Lx 10/25GbE LOM, Serial Number: N/A. A table lists the ports and their status:

Port	Permanent Address	Link State	Link Speed (Gbps)
1	7C8AE1D64166	up	25
2	7C8AE1D64167	up	1



Click here to see more information.

Server Configuration page

Use the **Server Configuration** → **Boot Options** page to configure the system boot mode and boot order.

The screenshot displays the XClarity Controller interface for a ThinkSystem SD630 V2 server. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration (highlighted), Adapters, Boot Options (highlighted), Server Properties, and BMC Configuration. The main content area is titled 'System Boot Mode and Boot Order' and includes a 'System name: SD630V2-4' header. It features radio buttons for 'UEFI Boot' (selected) and 'Legacy Boot'. Below this is the 'Boot Order' section, which shows 'Available devices' (USB Storage) and 'Boot order you want to configure' (VMware ESXi Hard Disk). The 'One Time Boot Device' section on the right allows selecting a device for one-time boot at the next server restart, currently set to 'No one-time boot'. Navigation arrows are visible on the left and right sides of the interface.

XClarity Controller

ThinkSystem SD630 V2 System name:SD630V2-4

Export SUPPORT 7:33 AM

System Boot Mode and Boot Order

☒ UEFI Boot
☐ Legacy Boot

Boot Order

Available devices:

USB Storage

Boot order you want to configure:

VMware ESXi
Hard Disk

LEGACY: OnBoard (0/17/0)
SSDSC2KB480G8L
02JG682D7A50581LEN
SATA PORT 0

LEGACY: OnBoard (0/17/0)
SSDSC2KB480G8L
02JG682D7A50581LEN
SATA PORT 1

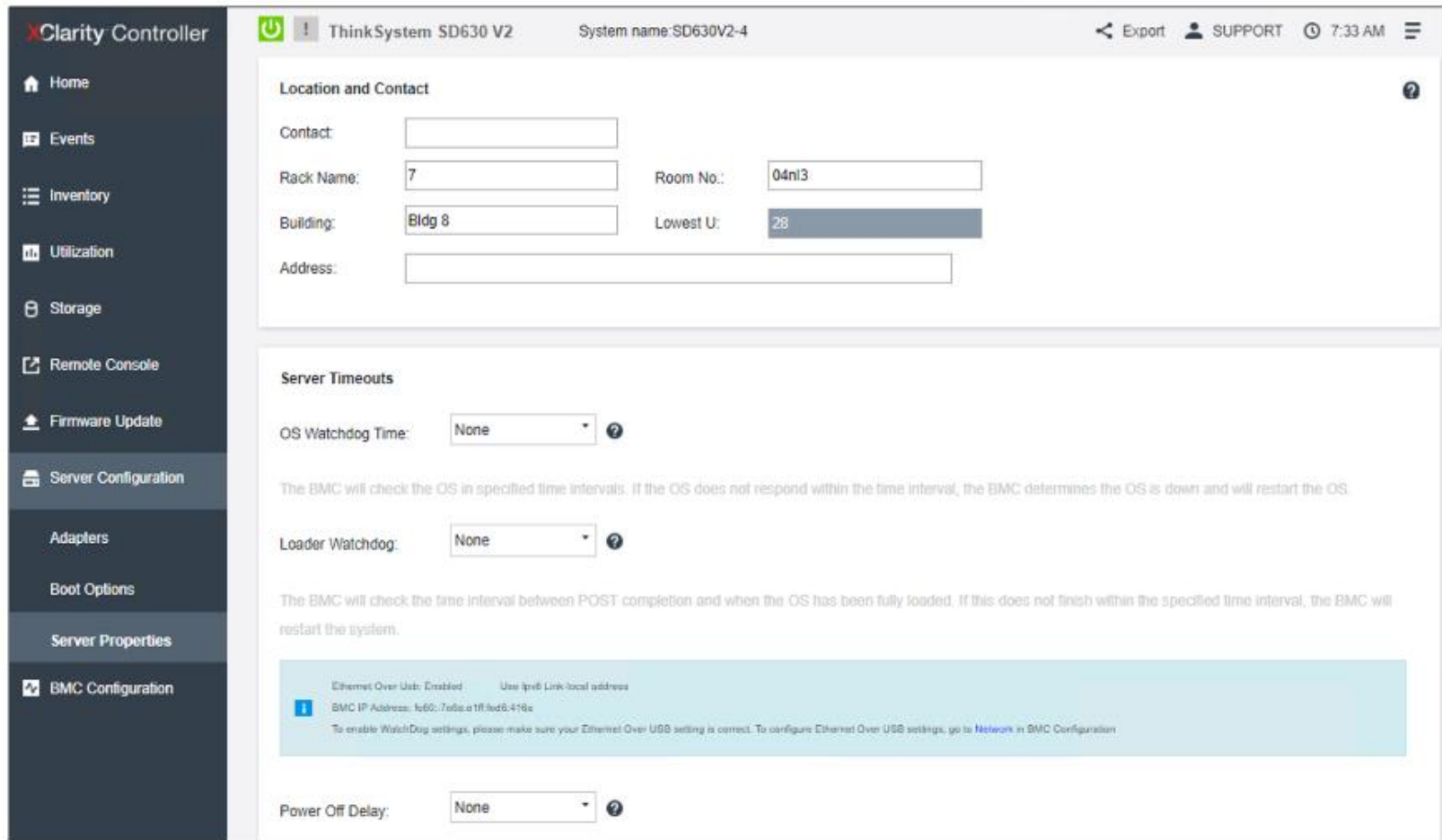
One Time Boot Device

Select the device for one-time boot at next server restart:

No one-time boot

Server Configuration page

The **Server Properties** page enables users to configure server location and contact information, select server timeout settings, and create a trespass message that will be displayed when a user logs in to XCC.



The screenshot displays the 'Server Configuration' page in the XClarity Controller interface. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration (highlighted), Adapters, Boot Options, Server Properties, and BMC Configuration. The main content area is titled 'ThinkSystem SD630 V2' with 'System name: SD630V2-4'. It features two sections: 'Location and Contact' and 'Server Timeouts'. The 'Location and Contact' section includes input fields for Contact, Rack Name (7), Room No. (04nl3), Building (Bldg 8), Lowest U. (28), and Address. The 'Server Timeouts' section includes dropdown menus for OS Watchdog Time (None) and Loader Watchdog (None), each with a help icon. A light blue informational banner at the bottom states: 'Ethernet Over USB: Enabled. Use IPv6 Link-local address. BMC IP Address: fe80::7a6a:21f:fed8:418e. To enable WatchDog settings, please make sure your Ethernet Over USB setting is correct. To configure Ethernet Over USB settings, go to Network in BMC Configuration.' The 'Power Off Delay' dropdown is also set to 'None'.

XClarity Controller ThinkSystem SD630 V2 System name: SD630V2-4 Export SUPPORT 7:33 AM

Location and Contact

Contact:

Rack Name: Room No.:

Building: Lowest U.:

Address:

Server Timeouts

OS Watchdog Time: ?

The BMC will check the OS in specified time intervals. If the OS does not respond within the time interval, the BMC determines the OS is down and will restart the OS.

Loader Watchdog: ?

The BMC will check the time interval between POST completion and when the OS has been fully loaded. If this does not finish within the specified time interval, the BMC will restart the system.

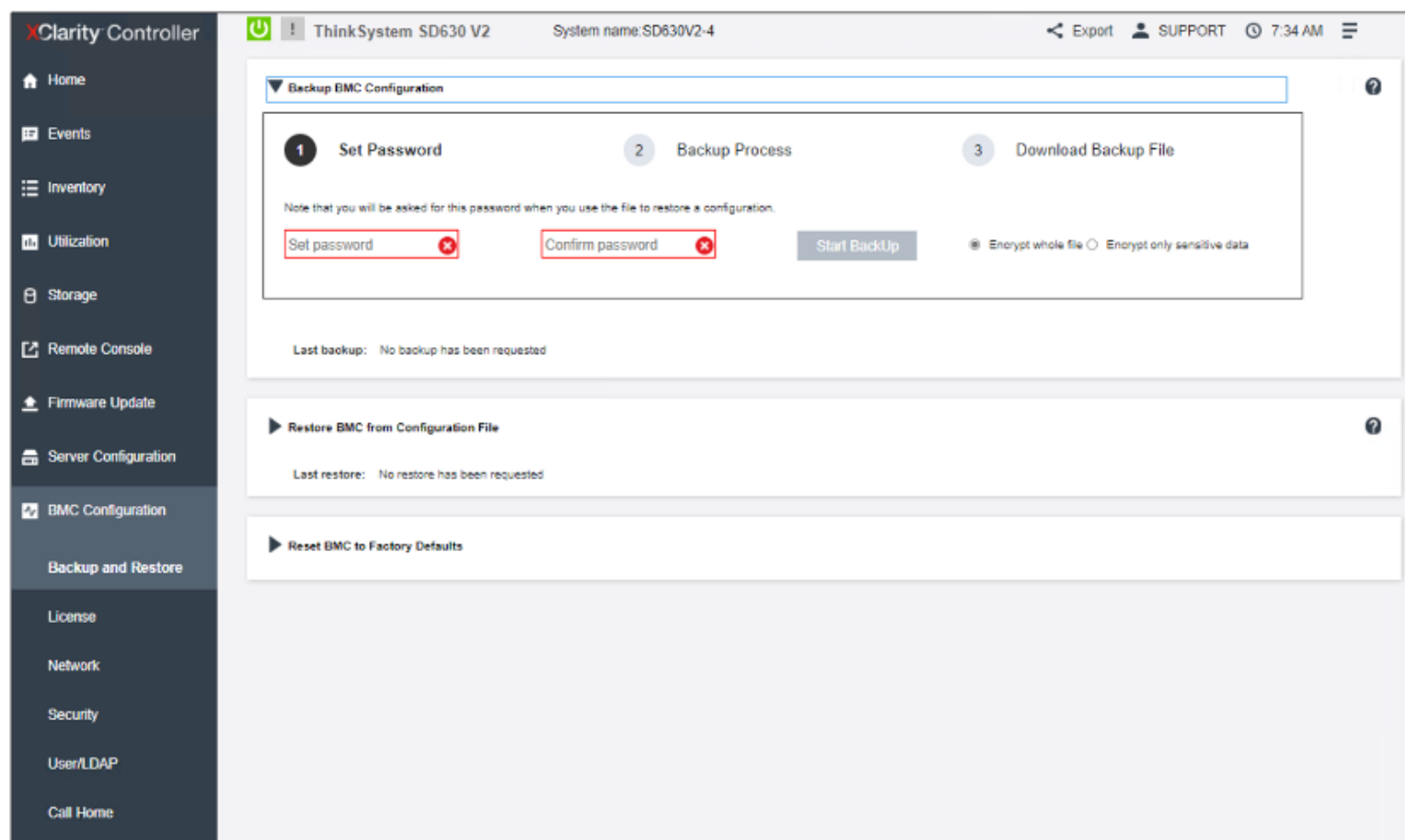
Information: Ethernet Over USB: Enabled. Use IPv6 Link-local address. BMC IP Address: fe80::7a6a:21f:fed8:418e. To enable WatchDog settings, please make sure your Ethernet Over USB setting is correct. To configure Ethernet Over USB settings, go to Network in BMC Configuration.

Power Off Delay: ?



BMC Configuration page

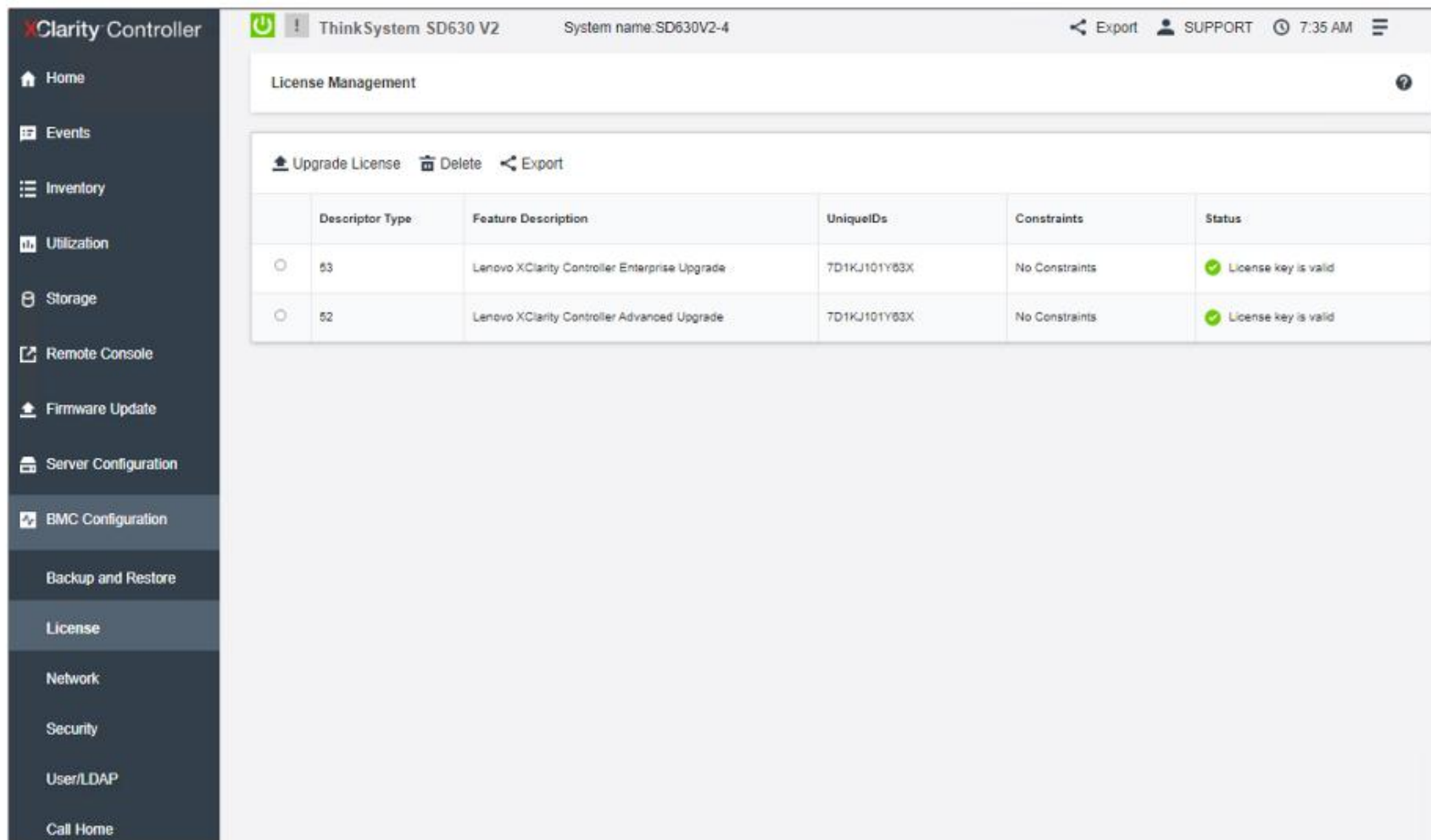
Use the **BMC Configuration** section to configure BMC settings. The **Backup and Restore** page allows users to reset the XCC configuration to factory defaults, and also to back up or restore configurations.



Click here to see more information.

BMC Configuration page

The **License** page allows users to manage XCC license activation keys.

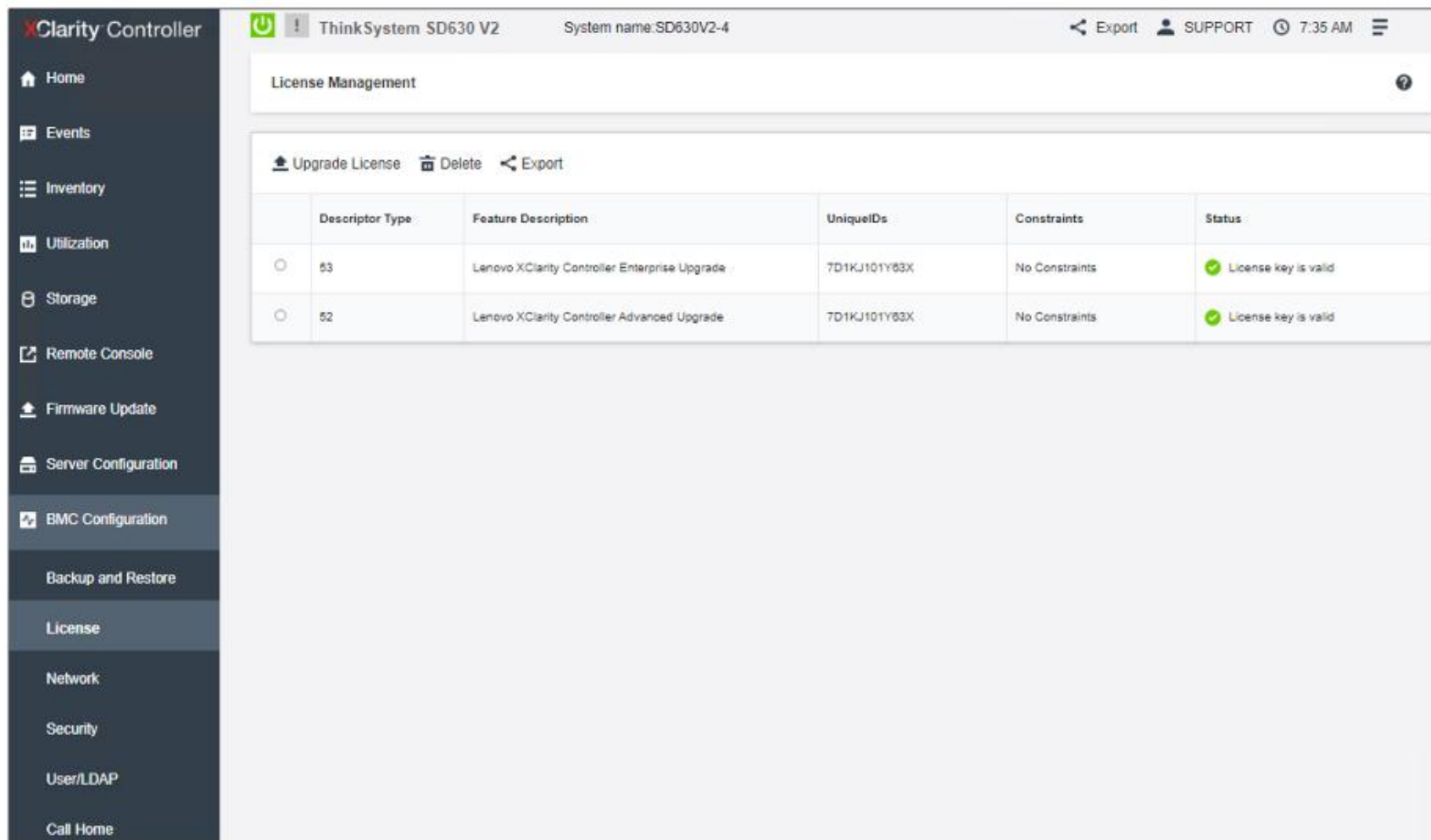


The screenshot displays the Lenovo XClarity Controller interface. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration, BMC Configuration (highlighted), Backup and Restore, License (highlighted), Network, Security, User/LDAP, and Call Home. The main content area is titled 'License Management' and includes a table with the following data:

	Descriptor Type	Feature Description	UniqueIDs	Constraints	Status
☐	53	Lenovo XClarity Controller Enterprise Upgrade	7D1KJ101Y63X	No Constraints	✔ License key is valid
☐	52	Lenovo XClarity Controller Advanced Upgrade	7D1KJ101Y63X	No Constraints	✔ License key is valid

BMC Configuration page

The **License** page allows users to manage XCC license activation keys.



The screenshot displays the Lenovo XClarity Controller interface. The left sidebar contains navigation links: Home, Events, Inventory, Utilization, Storage, Remote Console, Firmware Update, Server Configuration, BMC Configuration (highlighted), Backup and Restore, License (highlighted), Network, Security, User/LDAP, and Call Home. The main content area is titled 'License Management' and includes buttons for 'Upgrade License', 'Delete', and 'Export'. Below these buttons is a table with the following data:

	Descriptor Type	Feature Description	UniqueIDs	Constraints	Status
☐	53	Lenovo XClarity Controller Enterprise Upgrade	7D1KJ101Y63X	No Constraints	✔ License key is valid
☐	52	Lenovo XClarity Controller Advanced Upgrade	7D1KJ101Y63X	No Constraints	✔ License key is valid

BMC Configuration page

The **Network** page contains XCC networking property, status, and setting information.

The screenshot displays the 'Network' configuration page within the XClarity Controller interface. The left sidebar shows the navigation menu with 'BMC Configuration' and 'Network' highlighted. The main content area is divided into two sections: 'Ethernet Configuration' and 'DNS and DDNS'.

Ethernet Configuration

- Ethernet Ports:** Enabled (toggle switch).
- IP Settings Ethernet:**
 - Host Name:** SD630V2-4 (with a green checkmark).
 - ☐ Obtain Host Name from DHCP.
- IPv4:** Enabled (toggle switch).
 - Method:** Obtain IP from DHCP (dropdown menu).
 - Current IPv4 address:** 10.10.1.98
 - Network mask:** 255.255.252.0
 - Default gateway:** 10.10.0.3
- IPv6:** Enabled (toggle switch).

	IPv6 Address	Prefix
Link local IP:	fe80::7e8a:e1ff:fed6:4109	64
Stateless IP:		
Stateful IP:	::	0

DNS and DDNS

- DNS:** Auto-Enabled (toggle switch).
 - Preferred address type:** ☐ IPv4 ☒ IPv6
 - ☐ Use additional DNS address servers
 - ☒ Use DNS to discover Lenovo XClarity Administrator
- DDNS:** Enabled (toggle switch).
 - Method:** Use domain name obtained from the DHCP server*
 - Domain name:** syscoedlab.net

Quick Link

- Ethernet
- DNS and DDNS
- Ethernet over USB
- SNMP setup
- Service Port
- Address Restriction
- Front Panel USB

BMC Configuration page

The **User/LDAP** page provides access to XCC local user profiles and global login settings. It also enables users to configure LDAP security and certificate management for XCC.

XClarity Controller

Home

Events

Inventory

Utilization

Storage

Remote Console

Firmware Update

Server Configuration

BMC Configuration

Backup and Restore

License

Network

Security

User/LDAP

Call Home

ThinkSystem SD630 V2

System name:SD630V2-4

Export

SUPPORT

7:36 AM

Local User

LDAP

Allow logons from: Local first, then LDAP

1/12 local users

3/32 roles

Create

Global Settings

Name	Role	Advanced Attribute	Password Expiration	Active Sessions	Action
USERID	Administrator	Native	No Expiration		



BMC Configuration page

XCC supports the Call Home function on ThinkSystem V2 and later systems. Call Home allows users to create a service forwarder that will automatically send service data for any managed device to Lenovo Support when hardware error events are received from specific managed devices. This allows issues to be addressed more quickly.

Call Home is disabled by default. To enable Call Home, open XCC and select **BMC Configuration** → **Call Home**. Users then need to read and accept the terms and conditions description in XCC. Click [HERE](#) to see a screenshot.



XCC CLI

XCC supports the command-line interface (CLI). To access the XCC CLI, open an SSH session and enter the XCC IP address. Then, use the XCC username and password to log in to XCC.

For XCC CLI command syntax information and all available commands, refer to the following XCC CLI documentation site:

https://sysmgt.lenovofiles.com/help/index.jsp?topic=%2Fcom.lenovo.systems.management.xcc.amd.doc%2Fdw1lm_c_ch7_commandlineinterface.html&cp=3_1_10

- Introduction - Opening and Using the XClarity Controller - Configuring the XClarity Controller - Monitoring the Server Status - Configuring the Server - Configuring the Storage - Updating the firmware - License Management - Lenovo XClarity Controller REST API Command-line interface - IPMI Interface - Edge servers - Getting help and technical assistance - Lenovo XClarity Controller REST API - Lenovo XClarity Energy Manager - Lenovo XClarity Essentials	You can access the CLI through a SSH session. You must be authenticated by the XClarity Controller before you can issue any CLI commands. <u>Accessing the command-line interface</u> Use the information in this topic to access the CLI. <u>Logging in to the command-line session</u> Use the information in this topic to log in to the command line session. <u>Configuring serial-to-SSH redirection</u> This topic provides information about using the XClarity Controller as a serial terminal server. <u>Command syntax</u> Review the guidelines in this topic to understand how to enter commands in the CLI. <u>Features and limitations</u> This topic contains information about CLI features and limitations. <u>Alphabetical command listing</u> This topic contains a list of CLI commands in alphabetic order. Links are provided to topics for each command. Each command topic provides information about the command, its function,
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XCC helpful links

- XCC reference information on Lenovo Press:
<https://lenovopress.lenovo.com/lp0880-xcc-support-on-thinksystem-servers>
- XClarity Controller documentation:
https://sysmgt.lenovofiles.com/help/index.jsp?topic=%2Fxlcc_frontend%2Fxlcc_overview.html&cp=3
- Customer self support on GLOSSE – this page contains many links for ThinkSystem management tool articles and demo videos:
[https://glosse4lenovo.lenovo.com/wiki/glosse4lenovo/view/Customer%20Self%20Support/#Lenovo%20XClarity%20Controller%20\(LXCC\)](https://glosse4lenovo.lenovo.com/wiki/glosse4lenovo/view/Customer%20Self%20Support/#Lenovo%20XClarity%20Controller%20(LXCC))